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ABSTRACT

Recent measurements and simulations of wind farm wakes have highlighted interactions with the atmosphere that are not observed for wakes behind isolated turbines. This raises the question: How do wake recovery and its underlying physical mechanisms vary with wind farm size? In this work, we employ large eddy simulations to study the wake recovery behind 25 different-sized wind farms, ranging from an isolated turbine to an 81-turbine wind farm, in a deep conventionally neutral boundary layer. An analysis of the wake recovery mechanisms indicates a clear distinction between turbine-scale and wind farm-scale wake recovery physics. Turbine wake recovery is driven primarily by spanwise turbulent entrainment of energy, whereas recovery behind wind farms is governed by vertical turbulent entrainment and mechanical energy fluxes. The strong downward energy transport by the mean vertical flow highlights a fundamental difference with the wake development inside large wind farms, where this effect is limited. We further show that large wind farms can influence the atmosphere at high altitudes far downstream, even after the wind speed at hub height has mostly recovered. Finally, we evaluate how well several farm-level engineering models predict hub-height wind farm wake recovery and find that the overall trends are captured. Our findings significantly advance the understanding of wake recovery physics and can help to further improve the tools used for siting wind farms within clusters.

I. INTRODUCTION

The offshore wind energy sector is expected to see continued growth in the coming years, in response to the growing demand for renewable energy.[1,2] Both individual turbines and wind farms are increasing in scale, while the availability of the suitable shallow-water area to build them remains limited.[3] This development intensifies challenges in wind farm design and siting. For example, it is known that wake effects within wind farms can lead to significant power losses for downstream turbines.[4,5] Therefore, the wake recovery physics of turbine wakes within wind farms have been studied extensively.[6–8] However, recent studies have shown that wind farms form large aggregated wind farm wakes that can persist far downstream.[9–14] Consequently, wake interactions between wind farms are becoming more prominent, as increasingly more wind farms are projected to be built within relative proximity to each other.[15]

Direct observations have proven useful for gaining a general understanding of the scale and behavior of wind farm wakes. Christiansen and Hasager[16] were among the first to employ field measurements to observe wake effects further than 10 km downstream of a wind farm. Using satellite synthetic aperture radar, they studied the wakes behind the Horns Rev I and Rødsand I wind farms. They observed velocity deficits of 2% as far as 5 km downstream in unstable conditions and 20 km downstream in near-neutral conditions. Platis *et al.*[10] employed a research aircraft to measure the wakes in large wind farm clusters in the German North Sea and observed wake lengths ranging from 10 km in unstable conditions to 55 km in stable conditions. Further data analysis of the same measurement campaign revealed that wind farm density and turbine layout significantly influence wind farm wake characteristics.[17] Schneemann *et al.*[11] observed wind speed deficits of 21% at a distance of 55 km downstream of the DolWin2 cluster. The observed wake effects persisted for multiple hours a day. Many other authors[9,12,14,18,19] have also reported observations of wind farm wakes, often highlighting the role of atmospheric stability on their recovery.

To obtain a more fundamental understanding of how wind farm wakes recover and what impact they may have on neighboring farms, several authors have resorted to numerical simulations. Due to the large computational domains needed to study full wind farm wakes, these studies often make use of mesoscale[20–23] or Reynolds-averaged Navier–Stokes[24–26] simulations. However, such simulations do not resolve the relevant turbulent structures of the flow, and therefore only provide limited insight into the complex interactions between wind farms and the atmosphere. In contrast, large-eddy simulations (LES) enable more precise evaluation of the involved physics, as they use high temporal and spatial resolutions and resolve (rather than model) the relevant scales of flow turbulence. While the first LES of wind farms were highly idealized,[27] modern-day LES are able to account for a variety of atmospheric conditions, including thermal stratification,[28–35] Coriolis force,[36–38] dynamic wind direction changes,[39] and baroclinicity,[40,41] and may even be coupled to mesoscale models.[42]

Many LES studies have investigated the development of wind farm wakes and highlighted interactions with the atmospheric boundary layer (ABL) that are not observed for wakes of isolated turbines. For instance, Wu and Porté-Agel[33] used LES to investigate the role of free-atmosphere stratification in conventionally neutral boundary layers (CNBLs) on large wakes, and found velocity deficits of 3.5% at 10 km downstream of large 36-row wind farms in weakly stratified conditions. However, strong stratification was shown to induce standing gravity waves,[8,43] leading to a significant flow acceleration in the exit region of the wind farm, enhancing wake recovery. Maas and Raasch[35] performed an extensive LES study of (potential) wind farms in the German Bight area. They observed that wind farm wakes recover slower for denser farms and lower ABL heights. Furthermore, they highlight several flow effects only seen for large wind farms, including inversion layer displacement and an altered structure of the Ekman spiral. Moreover, they name the geostrophic forcing as an important recovery mechanism for wind farm wakes, in addition to turbulent entrainment. Stieren and Stevens[44] studied the impact of a 72-turbine wind farm on an equally large downstream farm. They observed significant power reductions in the entrance region of the waked wind farm. However, they found that this farm also experiences enhanced vertical entrainment fluxes, leading to faster wake recovery. Souaiby and Porté-Agel[13] performed LES of very large wind farms and found that both the length and layout of the wind farm had a significant impact on the wake recovery, and that the "infinite wind farm" regime was not yet reached for 36-row wind farms. Recent simulations by Lanzilao and Meyers[45] revealed that wind farm wake recovery is reduced in shallow ABLs due to limited vertical entrainment and highlighted a wake narrowing effect in the lateral direction that is not observed for isolated turbine wakes.

These larger-scale interactions with the atmosphere are often neglected in conventional wind turbine engineering models, which limits their ability to accurately predict the recovery of the large aggregated wakes that form behind wind farms. For instance, the far-wake recovery rate is often overpredicted by conventional turbine wake models.[13,46,47] Furthermore, effects such as wake deflection due to the Coriolis force[48] and the upward shift of the wake center due to non-uniform vertical entrainment[41,49] are generally not captured. Traditional multi-scale models, on the other hand, better capture large-scale interactions with the atmosphere, but typically only predict the flow within (infinite) wind farms, and do not provide information on the downstream wind farm wake recovery.[50–52]

Multiple designated wind farm engineering models attempt to overcome these shortcomings, to improve large-scale wind farm wake recovery predictions. The wake recovery model by Emeis[53] leverages the assumption that for an infinitely wide wind farm, the wind speed deficit is entirely compensated via vertical entrainment of energy, an established mechanism critical for wake recovery.[4,41] In 2020, Nygaard *et al.*[54] developed the TurbOPark model—a "top-hat" wind turbine model that addresses the typical overprediction of the recovery rate in the far wake. To this end, a non-linear wake expansion is considered, based on the prevailing atmospheric and wake-generated turbulence intensity (TI). Later, a Gaussian version of the model was developed as well.[55] Bastankhah *et al.*[48] proposed a semi-infinite wind farm model in 2024, that aims to provide improved far-wake recovery predictions by incorporating Coriolis effects, wind shear and wind veer in the ABL, which other models often neglect. The wake recovery is assumed to be driven by vertical entrainment, based on an atmospheric and a wind farm-specific contribution. That same year, Kasper and Stevens[41] proposed a simple farm-scale model to predict wind farm wake recovery, capturing the upward shift of the wake center based on the velocity shear in the ABL.

Despite these recent efforts, understanding and predicting wake recovery at the wind farm scale remains a major challenge, with several key questions still unresolved. In particular, it is unclear how wake properties and recovery dynamics vary with wind farm length and width, and to what extent existing engineering models accurately capture the recovery behind farms of different sizes. In this study, we address these knowledge gaps using LES of wind farms comprising 1–81 turbines, embedded in a deep CNBL. We perform an energy budget analysis over the wake region behind the wind farms, to investigate what physical mechanisms drive wake recovery. This analysis provides key insights into the fundamental differences between turbine-scale and wind farm-scale wake recovery, which may be leveraged to further improve wind farm siting tools. Additionally, we compare our LES results to several engineering models designed to predict wind farm wake recovery, to identify their strengths and weaknesses.

The remainder of this article is structured as follows: First, we discuss the methods we employed in Sec. II, including the LES framework and the engineering models. We then proceed to the results, starting with the characteristics of the simulated ABL in Sec. III, and the flow inside and around the wind farms in Sec. IV. Next, we analyze the mean wind farm wake properties. Section V evaluates the mean wind farm wake recovery and is followed by an analysis of the physical mechanisms that drive the recovery process in Sec. VI. Subsequently, Sec. VII compares the mean wake recovery predicted by the LES to several engineering models. Having discussed the mean wake properties, we then cover lateral and vertical variations in the wake structure in Sec. VIII. Finally, general conclusions are presented in Sec. IX.

II. METHOD

A. LES framework

1. Governing equations

We use an updated version of the code first developed by Albertson and Parlange[56] for our LES, which can simulate thermally stratified ABLs[57] and wind farm wakes[58] accurately. The governing equations are the incompressible equations for mass and momentum conservation and the transport equation for potential temperature

$$\partial_i \widetilde{u}_i = 0, \tag{1}$$

$$\partial_t \widetilde{u}_i + \partial_j(\widetilde{u}_i\widetilde{u}_j) = -\partial_i\widetilde{p}^* - \partial_j\tau_{ij} + \frac{g}{\theta_r}\delta_{i3}(\widetilde{\theta} - \theta_r) + \epsilon_{ij3}f_c(\widetilde{u}_j - U_j) + f_i, \tag{2}$$

$$\partial_t\widetilde{\theta} + \widetilde{u}_i\partial_i\widetilde{\theta} = -\partial_i q_i. \tag{3}$$

The notations (x_1, x_2, x_3), $(\widetilde{u}_1, \widetilde{u}_2, \widetilde{u}_3)$, and (x, y, z), $(\widetilde{u}, \widetilde{v}, \widetilde{w})$ are used interchangeably throughout the article. Here, $i = 1, 2, 3$ correspond to respectively streamwise, spanwise, and vertical directions, and tildes denote filtered variables. Furthermore, $\widetilde{u}_i$ and $\widetilde{\theta}$ are the resolved velocity and potential temperature fields, respectively. The geostrophic wind velocity is given by $U_i = -\epsilon_{ij3}\partial_j p_\infty/(\rho f_c)$, where ϵ_{ij3} is the alternating unit tensor, p_∞ is the mean pressure, ρ is the density, and f_c is the Coriolis frequency. The buoyancy term in Eq. (2) follows from the Boussinesq approximation, where θ_r is the reference temperature, g is gravity, δ_{ij} is the Kronecker delta, and $\Delta\theta = \widetilde{\theta} - \theta_r$. The deviatoric part of the sub-grid scale (SGS) tensor $\tau_{ij} = \tau^t_{ij} - (\tau^t_{kk}/3)\delta_{ij}$ and the SGS heat flux vector $q_i = \widetilde{u_i\theta} - \widetilde{u}_i\widetilde{\theta}$ are modeled using the AMD model.[59,60] The trace $\tau^t_{kk}/3$ of the total SGS tensor is absorbed into the modified pressure $\widetilde{p}^* = (\widetilde{p} - p_\infty)/\rho_0 + \tau^t_{kk}/3$.

The turbine force f_i is modeled using the actuator disk model.[27,61,62] The total turbine force is given by

$$F_t = \frac{1}{2}\rho C_T U_\infty^2 \frac{\pi}{4} D^2, \tag{4}$$

where C_T is the thrust coefficient and D is the rotor diameter. Since the free-stream velocity U_∞ is ill-defined within a wind farm, we use the approximation $C_T U_\infty^2 \approx C'_T U_D^2$ in Eq. (4). Here, U_D is the disk-averaged resolved velocity and $C'_T = C_T/(1-a)^2$ is the scaled thrust coefficient, where a is the induction factor. Details on the sampling of U_D from $\widetilde{u}_i$ and the projection of F_t onto the grid as f_i (including force filtering) can be found in Refs. 27, 29, 58, and 63. For convenience, tildes denoting LES filtering are omitted in the remainder of this article.

2. Boundary conditions and discretization

The simulation domain, displayed in Fig. 1, is horizontally periodic. At the top of the domain, a free-slip boundary with zero vertical velocity is enforced. We use the Monin–Obukhov similarity theory[64] to model the stresses at the ground, where the heat flux is set to zero, corresponding to neutrally stratified conditions. Second-order central finite differences and spectral differentiation are used to compute derivatives in vertical and horizontal directions, respectively. Time integration is performed using a third-order Adams–Bashforth scheme.[41,65] Furthermore, we employ a concurrent precursor method[66] to provide the inflow for the wind farm simulations, and use a Rayleigh damping layer[67] in the upper part of the domain to reduce the reflection of gravity waves.

3. Simulation setup and case overview

Next, we discuss how the simulations are configured. The simulation domain (refer to Fig. 1) measures $L_x \times L_y \times L_z = 52 \times 19.2 \times 4\,\text{km}^3$ in the streamwise, spanwise and vertical directions, respectively. The domain is discretized by a rectilinear grid, which is uniform in streamwise and spanwise directions, with grid resolutions of $\Delta_x = 50\,\text{m}$ and $\Delta_y = 20\,\text{m}$. The resolution in the vertical direction is $\Delta_z = 10\,\text{m}$ up to $z = 1.5\,\text{km}$, beyond which the grid is stretched progressively to reduce computational costs,[8,68] up to a maximum grid spacing of $\Delta_z \approx 120\,\text{m}$. We note that while the employed grid spacing is relatively coarse compared to previous studies,[40,41,69] the vertical resolution is well within the guidelines set by Wu and Porté-Agel[70] to properly resolve the turbine wakes. Furthermore, Appendix A validates that the horizontal grid resolution is sufficient to accurately capture the far-wake region behind individual turbines. The fringe region used to force the inflow spans the final $\Delta x_{\text{fr}} = 3\,\text{km}$ of the domain in the streamwise direction and $\Delta y_{\text{fr}} = 1\,\text{km}$ in the spanwise direction.

Each simulation contains one finite-sized staggered wind farm. We vary the number of turbine rows and columns in the wind farm from 1 to 9, with increments of 2. This results in a total of 25 cases, as listed in Table I. The smallest wind farm consists of an isolated turbine, while the largest wind farm comprises 9×9 turbines. We use constant streamwise and spanwise turbine spacings (in terms of D) of $s_x = 7$ and $s_y = 5$, respectively. The largest wind farm considered

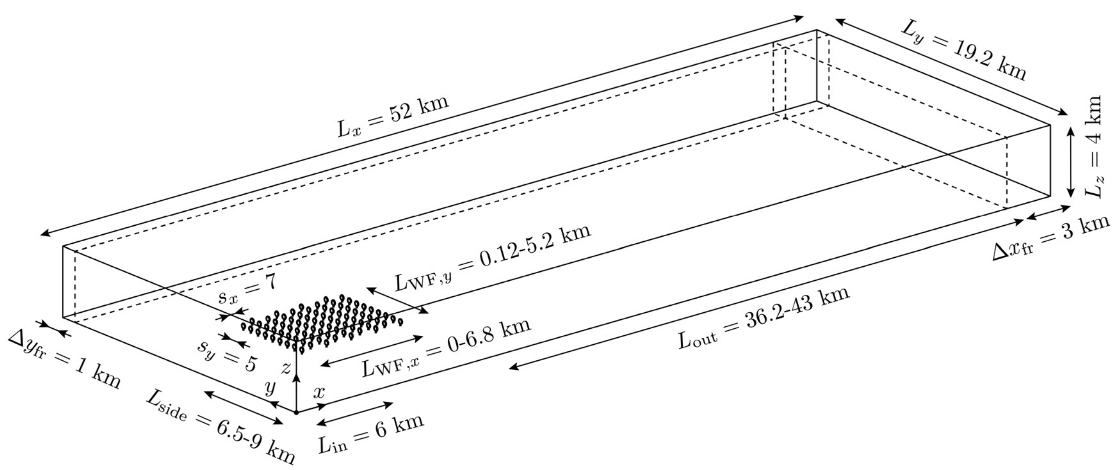

FIG. 1. Schematic illustration of the computational domain used in the LES. The illustration shows the 9×9 farm, the largest wind farm considered in this study. The streamwise and spanwise turbine spacings s_x and s_y are given in terms of rotor diameter D.

TABLE I. Overview of the 25 simulation cases, based on the number of turbine rows N_T and columns M_T in the wind farm ($N_T \times M_T$).

Cases	Increasing turbine columns $M_T \rightarrow$				
Increasing rows $N_T \downarrow$	1×1	1×3	1×5	1×7	1×9
	3×1	3×3	3×5	3×7	3×9
	5×1	5×3	5×5	5×7	5×9
	7×1	7×3	7×5	7×7	7×9
	9×1	9×3	9×5	9×7	9×9

covers roughly $L_{\mathrm{WF},x} \times L_{\mathrm{WF},y} = 6.8 \times 5.2\ \mathrm{km}^2$. We set a clearance of $L_{\mathrm{in}} = 6$ km upstream of the wind farm, which leaves a region of at least $L_{\mathrm{out}} \approx 36$ km behind each farm for the wake to develop. In spanwise direction the clearance is at least $L_{\mathrm{side}} = 6.5$ km, which is sufficiently wide, given that for deep neutral boundary layers (with an ABL height of $z_{\mathrm{ABL}} \approx 1$ km) confinement effects tend to be negligible for $L_{\mathrm{side}} \geq L_{\mathrm{WF},y}$.[43] The turbines have a diameter of $D = 120$ m, a hub height of $z_h = 90$ m, a thrust coefficient of $C_T = 0.75$, and an induction factor of $a = 0.25$. The actuator disks are automatically yawed to face the direction of the incoming flow.

We prescribe a geostrophic wind of $|U_i| = 12$ m/s, a Coriolis frequency of $f_c = 1.159 \cdot 10^{-4}\ \mathrm{s}^{-1}$, and a surface roughness height of $z_0 = 0.002$ m, corresponding to North Sea conditions.[32] This results in a hub-height velocity of $u_h = 10.6$ m/s. The direction of the geostrophic wind is controlled such that at hub height z_h the flow is in x direction.[71] The initial temperature profile contains a neutral layer with $\theta = 286$ K that reaches up to $z_{\mathrm{ABL}} = 1$ km, capped by an inversion of 3 K over 200 m, above which there is a free-atmosphere stratification of 5 K/km. In the lower part of the ABL, random perturbations are added to the initial condition to spin up turbulence.

The simulations are performed in two stages. First a spin-up simulation is run to develop the ABL to a quasi-steady state. After 7 h, the simulation continues as a precursor-successor simulation, and the turbines forces are activated. After a 1-h adjustment period, statistics are collected over 3 h.

B. Engineering models

This section provides an overview of the different wake recovery models that are compared to the LES in Sec. VII, ordered by their resolution, from farm-scale to turbine-scale. For brevity, only short conceptual descriptions are provided here, with the key model differences summarized in Table II. Extended mathematical definitions are presented in Appendix B.

1. Emeis (2010) model

The Emeis (2010) model[17,53] is a farm-scale wake model that provides a simple and easy-to-use estimation for large-scale wake recovery. For an infinitely wide wind farm, this model assumes the wind speed deficit $\delta(x') = u_h - u$ behind the wind farm is entirely compensated by a vertical momentum flux τ_{ver}. Here, x' is distance behind the wind farm. Furthermore, τ_{ver} is estimated using the internal boundary layer (IBL) height z_{IBL} and friction velocity u_*, which are considered input to the model. The initial velocity deficit δ_0 immediately behind the wind farm is an input parameter as well. The model uses no empirically-determined or tuning constants.

TABLE II. Comparison of dedicated wind farm wake recovery models based on resolution, tuning constants and input parameters. For brevity, input parameters that solely depend on wind turbine and farm geometry are not listed.

Model	Resolution	Nr. of tuning constants	(Atmospheric) input parameters
Emeis (2010)	Farm-scale	0	u_*, δ_0, z_{IBL}
Kasper (2024)	Farm-scale	2	$u_{\mathrm{ABL}}(z)$, $I_{\mathrm{ABL}}(z)$, δ_0, z_{IBL}
Bastankhah (2024)	Row-scale	3	u_*, f_c, z_0, z_{ABL}
TurbOPark	Turbine-scale	3	$I_{\mathrm{ABL}}(z = z_h)$

2. Kasper (2024) model

The Kasper (2024) model[41] is a farm-scale wake model that leverages insights into the prevailing atmospheric conditions to predict the wake recovery behind large wind farms. The velocity deficit profile $\delta(z, x')$ inside the wake is assumed to be laterally uniform, and Gaussian-shaped in vertical direction. The Kasper (2024) model aims to provide accurate far-field recovery predictions, by accounting for displacement of the wake profile in addition to wake expansion. These mechanisms are modeled empirically, using the velocity and turbulence intensity profiles $u_{\mathrm{ABL}}(z)$ and $I_{\mathrm{ABL}}(z)$, which are input parameters to the model, and two tuned constants. Similarly to the Emeis (2010) model, z_{IBL} and δ_0 must be known or estimated for the considered wind farm.

3. Bastankhah (2024) model

The Bastankhah (2024) model[48] is a physics-based wake model designed to predict the flow inside and behind semi-infinite wind farms. It considers five different mechanisms to determine the local velocity deficit at hub height: turbine forces, wake recovery, Coriolis effects, wind shear, and wind veer. Using various approximations for the different terms of the momentum equation, a formulation is found for the laterally averaged flow at hub height. Due to the inclusion of Coriolis effects and wind veer, the model captures both the streamwise deficit $\delta_U(x')$ and the wake deflection $\delta_V(x')$. The Bastankhah (2024) model computes the velocity deficit contributions behind each turbine row individually, and then sums over all rows to obtain the total flow deficit. To this end, this model uses several empirical constants, the thrust coefficient C_T, and the atmospheric properties u_*, z_0, z_{ABL}, and f_c.

4. TurbOPark model

The TurbOPark model[54] is an established turbine-scale wake model designed to deliver improved far-wake velocity predictions compared to traditional turbine wake models. This model assumes a non-linear turbine wake expansion based on the local turbulence intensity I_{loc}, which contains an atmospheric contribution ($I_{\mathrm{ABL}}(z = z_h)$) and a wake-generated contribution I_w. The wake is modeled laterally and vertically uniform (also known as a top-hat profile), and the velocity deficit behind each turbine is computed individually, using C_T, $I_{\rm loc}$, and three empirical constants. The contributions of all turbines are then summed in quadrature to obtain the total deficit $\delta(x, y, z)$.

III. BOUNDARY LAYER CHARACTERISTICS

We proceed to the simulation results, starting with the properties of the undisturbed ABL, sampled from the precursor domain. Figure 2(a) shows the horizontal wind velocity $u_{\rm ABL} = \langle\sqrt{\overline{u}^2 + \overline{v}^2}\rangle$, where $\bar{\cdot}$ and $\langle . \rangle$ denote temporal and spatial averaging, respectively. The velocity at hub height equals $u_h = 10.6$ m/s and $u_{\rm ABL}$ approaches $|U_i|$ above the inversion height of 1 km, which makes for a relatively deep ABL. We find that the wind shear in the surface layer corresponds to a Hellmann exponent of roughly 0.11, consistent with open sea conditions.[72] The mean wind angle $\alpha = \langle \tan^{-1}(\overline{v}/\overline{u})\rangle$, shown in Fig. 2(b), equals 0° at hub height (by design) and moves toward roughly −13° in the free atmosphere. The veer over the rotor disk is relatively small at roughly 1.6°. Figure 2(c) displays the mean turbulent stress, given by

$$\tau = \left\langle \sqrt{\left(\overline{u'w'}\right)^2 + \left(\overline{v'w'}\right)^2} \right\rangle, \tag{5}$$

which decreases with increasing height within the ABL. Note that the SGS-components are included in τ, e.g., via $\overline{u'w'} = \overline{uw} - \overline{u} \cdot \overline{w} + \overline{\tau}_{xz}$. The measured friction velocity equals roughly $u_* \approx 0.38$ m/s in all simulations. Furthermore, the horizontal turbulence intensity, which we define as

$$I_{\rm ABL} = \left\langle \sqrt{\overline{u'^2} + \overline{v'^2}} / \sqrt{\overline{u}^2 + \overline{v}^2} \right\rangle, \tag{6}$$

equals 8.7% at hub height. Figure 2(d) shows the potential temperature profile with height z. Given that the simulated ABL is conventionally neutral, $\langle\overline{\theta}\rangle$ remains constant within the boundary layer. Both the inversion layer starting at $z_{\rm ABL} = 1$ km and the stably-stratified free atmosphere above have a clear impact on the profile.

IV. FLOW INSIDE AND AROUND THE WIND FARMS

Next, we evaluate the flow through the wind farms. Figure 3 shows the instantaneous horizontal velocity fields $\sqrt{u^2 + v^2}/u_h$ for six simulations with different wind farms. Note that the figures in this work use the shifted streamwise coordinate $x' = x - x_{\rm end}$, where $x_{\rm end}$ is the location "immediately behind" the wind farm where the wake deficit is strongest, i.e., at a distance of $2D$ behind the rearmost turbine row.[73] This convention enables easier comparison of the wakes behind the wind farms, which is our main focus. In each simulation, meandering turbine wakes are clearly visible as regions of reduced velocity downstream of the turbine locations (indicated by the black lines). In the single-row simulations shown in Figs. 3(a) and 3(d), the individual turbine wakes recover relatively quickly and remain isolated from each other. The addition of 2 extra turbine rows in the simulations of Figs. 3(b) and 3(e) result in wake interactions, as wakes of upstream turbines travel downstream and encounter subsequent turbines (and wakes). As a consequence, higher velocity reductions are observed within the wind farm and a more unified wake trails behind it. Even larger velocity deficits are found in the 9-row setups of Figs. 3(c) and 3(f). There, the turbine wakes expand and merge significantly within the wind farm, and consequently an aggregated wind farm wake is formed behind it. The number of turbine columns M_T has less impact on the flow through the wind farms than the number of rows N_T, though the wakes appear to persist further downstream for wider wind farms.

A more quantitative assessment can be made by evaluating the temporally- and spanwise-averaged velocity $u_s = \langle\sqrt{\overline{u}^2 + \overline{v}^2}\rangle/u_h$ at hub height as a function of x', shown in Fig. 4(a). It should be noted that the exact definition of the spanwise averaging region can affect the interpretation of the results, as explained in Appendix C. Here, we consider a fixed width of s_yD per turbine to determine the averaging region (e.g., for case 1 × 3 the spanwise extent equals $3s_yD$), to avoid a strong non-linear scaling with M_T. In each simulation displayed in Fig. 4(a), we observe the characteristic overall development of the flow velocity throughout the wind farm. Upstream of the farm, there is a flow deceleration due to the blockage effect.[74,75] Progressing downstream, sudden drops in flow speed occur at the turbine locations as energy is extracted from the flow, followed by gradual wake recovery. Behind the wind farm, the accumulated wake recovers slowly toward the undisturbed flow condition.

The influence of N_T on the flow through the different wind farms is as expected. Extra rows result in additional flow speed reductions and, consequently, a lower wind farm exit velocity $u_{\rm end} = u_s(x' = 0)$ is observed. As N_T increases, the flow inside the wind farm further

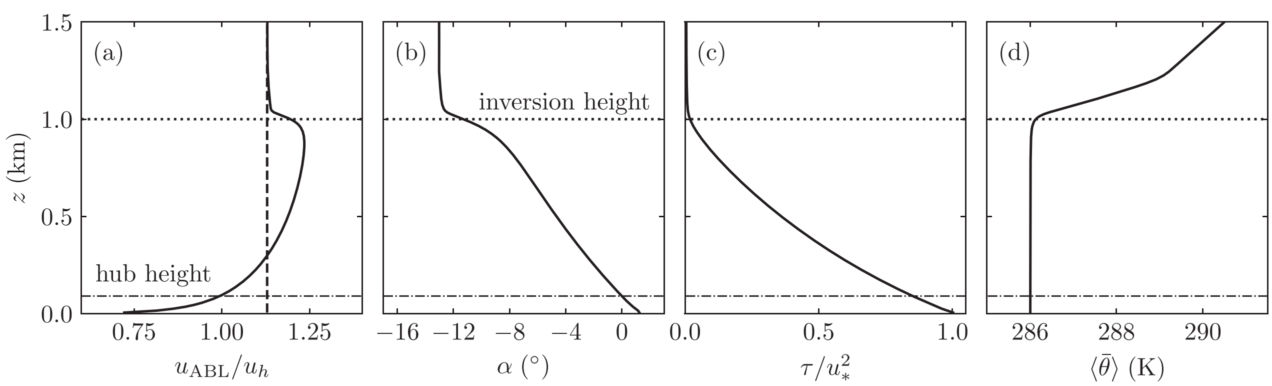

FIG. 2. The averaged properties of the ABL as a function of height z. (a) The horizontal velocity magnitude $u_{\rm ABL}$, normalized by the hub-height velocity u_h. The imposed geostrophic wind is shown by the vertical dashed line. (b) The wind angle α, (c) the shear stress τ, normalized by the friction velocity u_*, and (d) the potential temperature profile $\langle\overline{\theta}\rangle$.

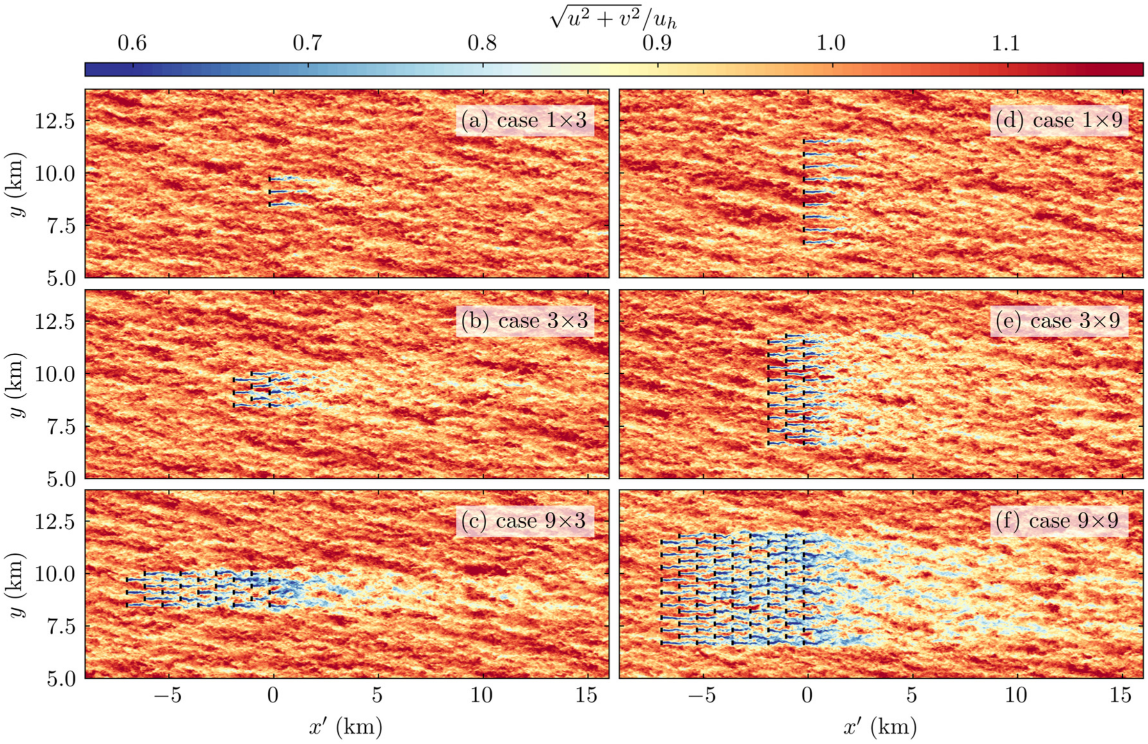

FIG. 3. Instantaneous horizontal velocity magnitude fields in the $x'y$-plane at hub height, for (a)–(c) three of the 3-column wind farms and (d)–(f) three of the 9-column wind farms. Black lines indicate the turbine positions. Note that only part of the simulation domains is shown; for the full domain size, refer to Fig. 1.

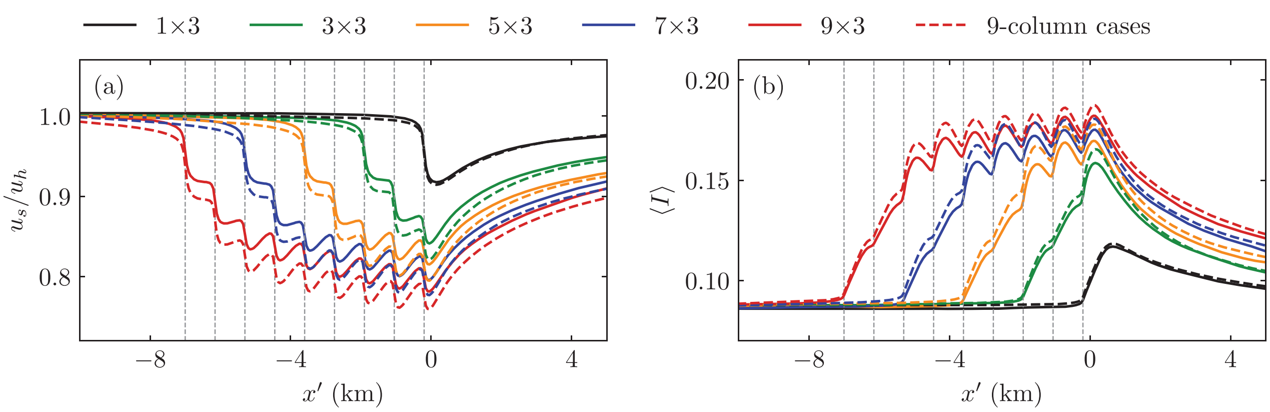

FIG. 4. Temporally- and spanwise-averaged properties inside the wind farm as a function of distance x', at hub height, for the $M_T = 3$ cases (solid lines) and $M_T = 9$ cases (dashed lines). (a) The velocity magnitude u_s, and (b) the turbulence intensity $\langle I \rangle$. The vertical gray dashed lines indicate the locations of the turbine rows.

approaches a fully developed regime,[33] such that the contribution of additional rows to u_{end} decreases progressively. Regarding the influence of M_T, we find that a wind farm's width significantly affects the flow upstream. The wider farms exhibit enhanced flow blockage, and this effect increases for longer farms. The reduction in mean flow velocity persists throughout the wind farm, resulting in lower farm exit velocities for larger M_T.

The trends in the mean turbulence intensity $\langle I \rangle$, shown in Fig. 4(b), are similar to those in u_s. We find that $\langle I \rangle$ builds up rapidly throughout the first few turbine rows and stagnates further downstream in the wind farm. Furthermore, the total wake-generated turbulence intensity increases with both N_T and M_T. However, in contrast to u_s, there is no appreciable effect of wind farm size on $\langle I \rangle$ directly upstream of the wind farm.

Additionally, wind farm size impacts the development of the internal boundary layer. Figure 5 displays the temporally-averaged velocity field $\sqrt{\overline{u}^2+\overline{v}^2}$ in the $x'z$-plane at the middle turbine column, for all 9-column wind farms. The white dashed lines depict the height

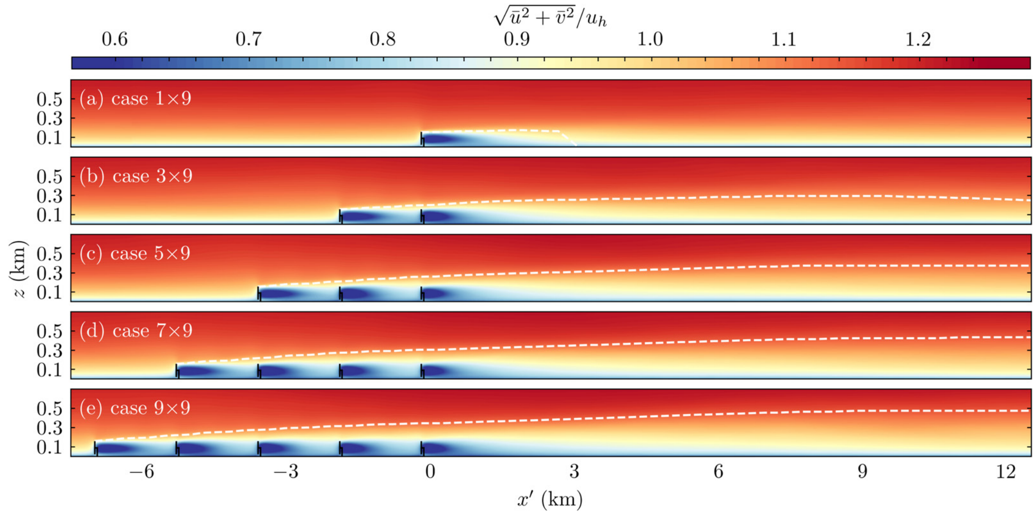

FIG. 5. Temporally-averaged velocity magnitude $\sqrt{\bar{u}^2+\bar{v}^2}$ in the $x'z$-plane, at the middle turbine column, for the 9-column cases. The white dashed line denotes the IBL height z_{IBL}. Note that only part of the simulation domain is shown, and that the vertical extent is stretched by a factor of two to improve clarity.

of the internal boundary layer z_{IBL}, which we define as the height where the temporally- and spanwise-averaged flow first recovers to 97% of the undisturbed wind speed at that height.[8,76] For the 1×9 case, the wake-generated turbulence is weak and development of the IBL is limited. The IBL decays entirely within a few kilometers downstream, and its height z_{IBL} barely exceeds the top of the rotor (at $z_h + D/2 = 150$ m). The addition of two extra turbine rows enhances the development of the IBL significantly, as shown in Fig. 5(b). Interestingly, the majority of the IBL growth in the 3×9 simulation occurs behind the wind farm. We observe that for the longer wind farms, the IBL develops progressively more within the farm, due to increased velocity shear and turbulence [refer to Figs. 4(a) and 4(b)]. Nonetheless, the IBL keeps growing behind the wind farm over a distance significantly longer than the length of the wind farm itself. Of course, the relatively deep ABL height and high turbulence intensity characteristic of conventionally neutral conditions accommodate continued development of the IBL.[76] Very far downstream, the IBL first decreases in height, before the flow ultimately approaches undisturbed conditions again.

V. WIND FARM WAKE RECOVERY

In this section, we evaluate the mean wake recovery behind the wind farms, and find it is influenced by both wind farm length and width. Figure 6(a) compares the mean velocity u_s/u_h behind the wind farm at hub height, between farms with a different number of turbine rows N_T. Similarly, Fig. 6(d) shows u_s/u_h for wind farms with varying numbers of turbine columns M_T. For each wind farm, we observe that the wake flow is characterized by a relatively fast near-farm recovery rate followed by a slower recovery rate downstream. Generally, the wind farm exit velocity u_{end} decreases with increasing N_T and M_T, causing the velocity deficit to remain significant over larger distances for the longer and wider farms. The addition of extra rows slows down the wake recovery even for large values of N_T, in agreement with simulations by Souaiby and Porté-Agel[13] In contrast, we find that after the transition from $M_T = 1$ to $M_T = 3$, the effect of additional columns is minimal.

Different representations of the data of Figs. 6(a) and 6(d) yield additional insights into the scaling of the near-farm and far-field recovery. For instance, normalizing by u_{end}, as shown in Figs. 6(b) and 6(e), reveals that near-farm recovery rates overlap at $x' = 0$ for the $N_T \geq 3$ and $M_T \geq 3$ wind farms. Further downstream, the curves deviate, as the undisturbed flow condition varies between simulations using this normalization. Figures 6(c) and 6(f) show the flow velocity as a function of $x' - L_{\mathrm{wake}}$, where L_{wake} is the wake length. Here, we adopt the wake length definition used by Cañadillas *et al.*,[12] who determine L_{wake} as the distance behind the wind farm where the mean flow at hub height first reaches 95% of the undisturbed flow velocity (though we note this threshold is somewhat arbitrary). We observe that the curves overlap in the far-field for all farms with $N_T \geq 3$ and $M_T \geq 3$, and only deviate in the region immediately downstream of each wind farm. These different representations of the data demonstrate that the near-farm recovery is highly dependent on wind farm size and shape, while the far-field recovery is independent of the specific wind farm configuration. The single-row and single-column cases in Fig. 6 do not follow these trends observed for the larger wind farms. This deviation relates to the physical mechanisms driving the wake recovery, discussed in detail in Sec. VI, which are found to be similar for wind farms of at least three rows and three columns.

We now explore how the wake length L_{wake} varies with wind farm length and width. Figures 7(a) and 7(b) depict L_{wake} as a function of turbine rows N_T and turbine columns M_T, respectively. Figure 7(a) shows that the wake length increases with N_T for all wind farm widths considered. This result is expected, because more energy is extracted from the flow for larger N_T. As the number of rows increases, the growth rate of L_{wake} with N_T reduces, though no plateau is reached within our set of simulations. For the wake length as a function of M_T, shown in Fig. 7(b), such a plateau does appear to be reached around $M_T = 7$. For $N_T \geq 3$, the wake length grows rapidly with M_T initially, while stagnating at larger values. Thus, wind farm width only has a

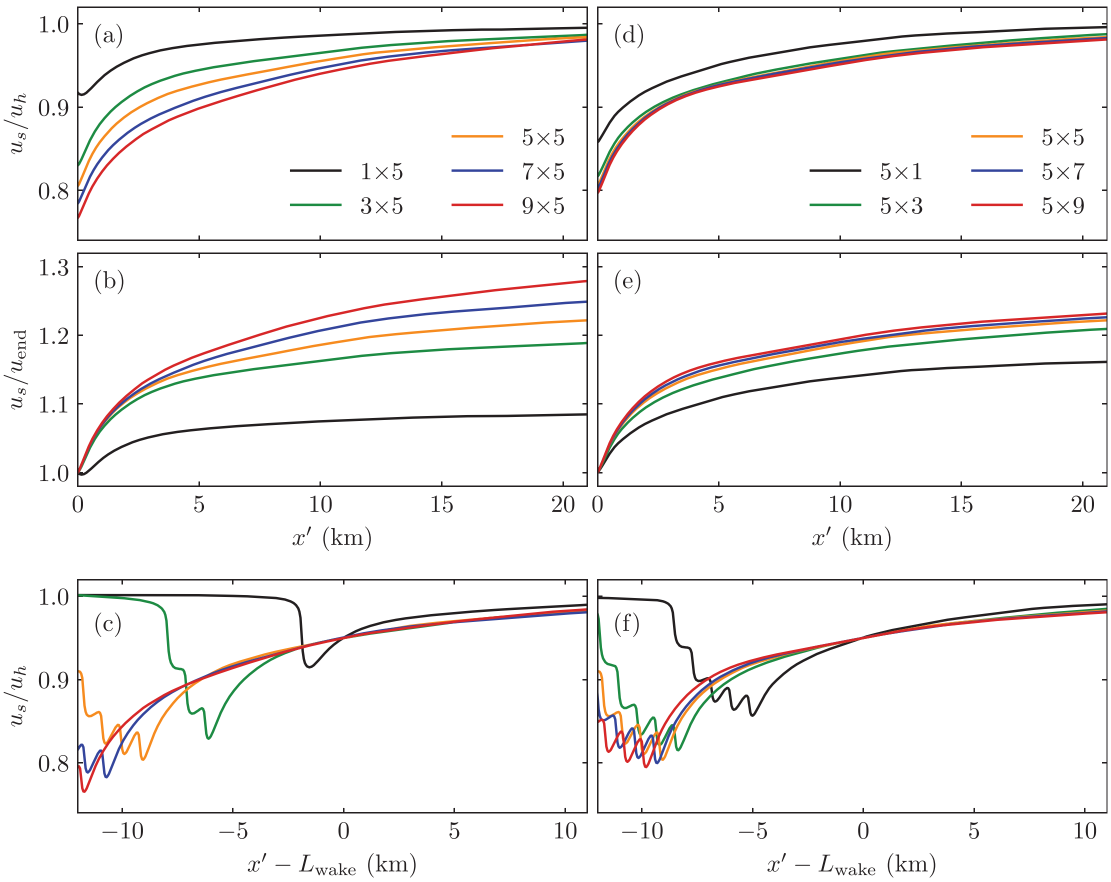

FIG. 6. Mean velocity development downstream of the wind farms as a function of distance x', using various normalizations and offsets. Comparison between (a)–(c) different number of turbine rows N_T, and (d)–(f) different number of turbine columns M_T. (a) and (d) Velocity normalized by hub-height velocity u_h. (b) and (e) Velocity normalized by wind farm exit velocity u_{end}. (c)–(f) Distance offset by the wake length L_{wake}.

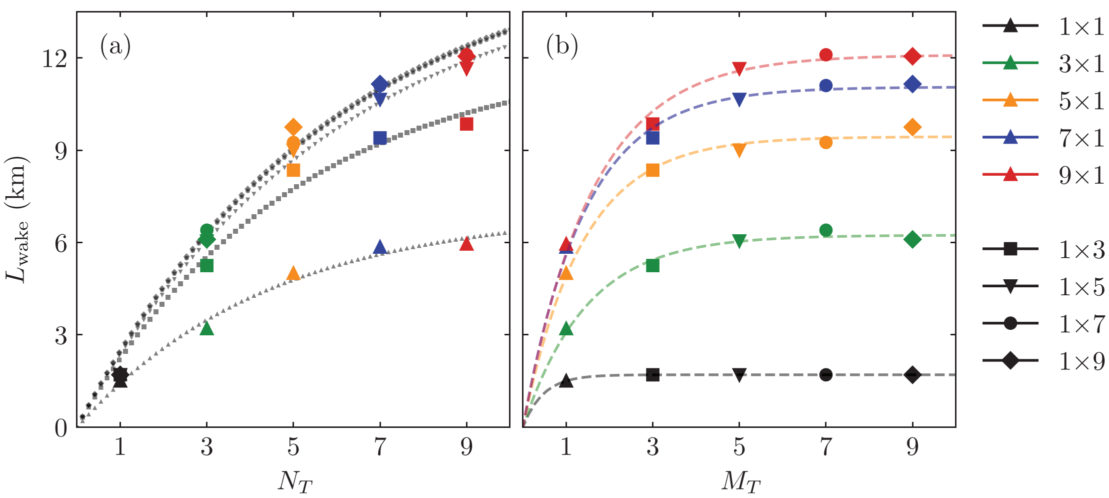

FIG. 7. Comparison of the wind farm wake length L_{wake}, as a function of (a) turbine rows N_T, and (b) turbine columns M_T in the wind farm ($N_T \times M_T$). Different colors correspond to farms with different N_T, while different markers represent farms with different M_T. Least-squares fitted asymptotic exponential decay functions, as described by Eq. (7), are plotted through the data points in broken lines.

limited impact on L_{wake}, which becomes negligible for wide ($M_T \geq 7$) wind farms. In the single-row simulations no aggregated wind farm wake is formed, and as a result L_{wake} is nearly independent of wind farm width.

Next, we attempt to identify a simple functional relationship between L_{wake} and the length or width of the wind farm. We postulate this function should exhibit some key characteristics. The function must equal 0 for $N_T = 0$ (or $M_T = 0$), and monotonically increase with the number of turbines, as each turbine extracts additional energy from the flow. Furthermore, the function should have a positive but monotonically decreasing derivative, which vanishes for very large numbers of turbines. These conditions reflect an asymptotic trend toward a fully developed state in which the wind farm is in equilibrium with the ABL, found in simulations of "infinitely large" (periodic) wind farms.[6] A simple function that fits these requirements is the asymptotic exponential decay function, which has the form

$$f(x) = a[1 - \exp(-bx)], \tag{7}$$

where a and b are constants. Least-squares fits using this form are displayed in Figs. 7(a) and 7(b), where x is taken to be the number of turbine rows N_T or columns M_T, respectively. We find the fitted curves excellently match with the LES measurements. However, the obtained constants a and b, documented in Appendix D, do not appear to be trivially dependent on M_T and N_T. Likely, a and b depend significantly on the atmospheric conditions and wind farm properties, and their prediction *a priori* remains a topic for future research. Nevertheless, Eq. (7) may be useful in practical applications, allowing easy extrapolation of measured wake properties to wind farms of different sizes.

VI. WAKE RECOVERY MECHANISMS

To improve our understanding of how turbine-scale and wind farm-scale wake recovery physics differ, we perform a total kinetic energy budget analysis over the wake region behind the wind farms. The analysis draws inspiration from the typical energy budget analyses often used to study wind turbine and farm power generation,[6–8,41,77,78] and the wake momentum budget analysis of Ref. 48. First, we consider a control volume behind the wind farm to define the wake region, as depicted in Fig. 8(a). This volume has a cross-sectional area A_{cs} of size $L_{\text{WF,y}} \times D$, and a length L_{wake} (as defined in Sec. V). Next, we multiply Eq. (2) by u_i, average over time, and integrate over the wake cross section A_{cs}. After rearranging the terms, this yields the equation

$$\begin{aligned}
&\underbrace{-\int_{A_{\text{cs}}} \partial_1 \left[\overline{u}_1 \left(\frac{1}{2}\overline{u}_i\overline{u}_i + \frac{1}{2}\overline{u_i'u_i'} + \overline{p} \right) \right] \mathrm{d}A}_{\delta\mathbb{M}_x,\text{Mechanical energy flux (streamwise)}} \\
&= \underbrace{\int_{A_{\text{cs}}} \partial_j \left[\left(\overline{u}_2\delta_{j2} + \overline{u}_3\delta_{j3} \right) \left(\frac{1}{2}\overline{u}_i\overline{u}_i + \frac{1}{2}\overline{u_i'u_i'} + \overline{p} \right) \right] \mathrm{d}A}_{\delta\mathbb{M}_{yz},\text{Mechanical energy flux(lateral/vertical)}} \\
&+ \underbrace{\int_{A_{\text{cs}}} \partial_j \left(\frac{1}{2}\overline{u_i'u_i'u_j'} + \overline{u}_i\overline{u_i'u_j'} + \overline{u_i\tau_{ij}} + \overline{u_j'p'} \right) \mathrm{d}A}_{\delta\mathbb{T}_t,\text{Total turbulent transport}} \\
&+ \underbrace{\int_{A_{\text{cs}}} \epsilon_{ij3} f_c \overline{u}_i U_j \mathrm{d}A}_{\delta\mathbb{G},\text{Geostrophic forcing}} + \underbrace{\int_{A_{\text{cs}}} -\overline{\tau_{ij}S_{ij}}\mathrm{d}A}_{\delta\mathbb{D},\text{ Dissipation}} \\
&+ \underbrace{\int_{A_{\text{cs}}} \frac{g}{\theta_r}\delta_{i3}\left(\overline{u}_i\theta_r - \overline{u_i\theta} \right) \mathrm{d}A}_{\delta\mathbb{B},\text{ Buoyancy}},
\end{aligned} \tag{8}$$

which still depends on x'. Here, $\delta\mathbb{M}_x$ and $\delta\mathbb{M}_{yz}$ represent the change in the streamwise and orthogonal mechanical energy fluxes, respectively. Furthermore, $\delta\mathbb{T}_t$ represents the total turbulent transport, $\delta\mathbb{G}$ the geostrophic forcing, $\delta\mathbb{B}$ the turbulence production/destruction due to buoyancy, and $\delta\mathbb{D}$ the turbulent dissipation, where S_{ij} is the strain-rate tensor. The turbine force contribution has been omitted from Eq. (8), since it equals zero in the wake region. Moreover, we use $\delta\mathbb{R}$ to denote the budget residual (the sum of all terms, representing numerical error). We note that in the following analysis, the energy

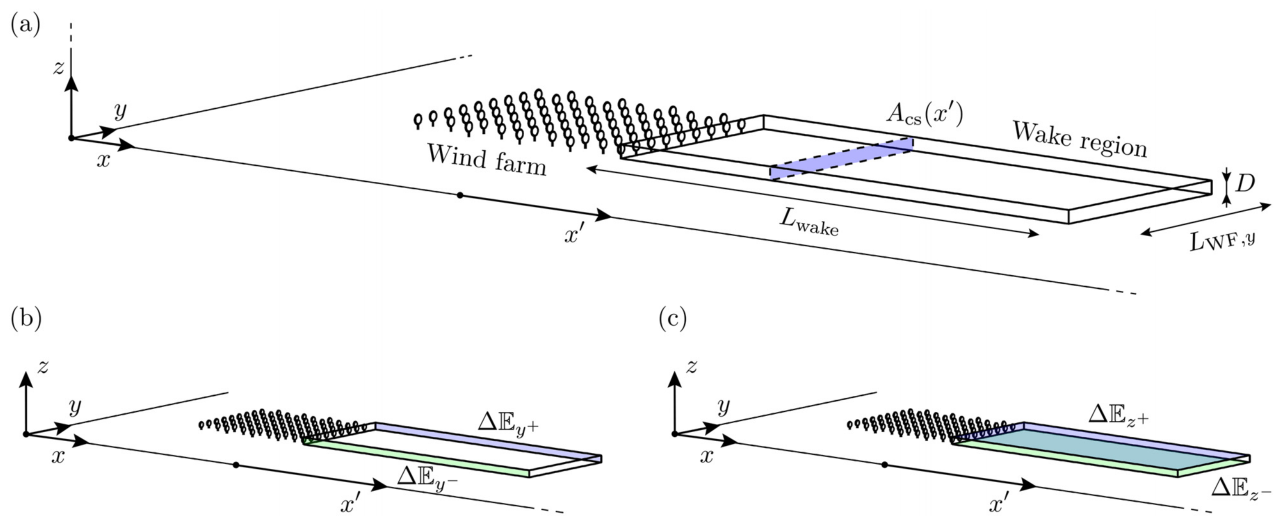

FIG. 8. Schematic of the energy budget analysis of the wake region behind the wind farms. (a) Illustration of the control volume. (b) and (c) The mean flux densities $\Delta\mathbb{E}$ through the sides and top/bottom of the wake region, respectively. The schematic depicts the 9×9 wind farm.

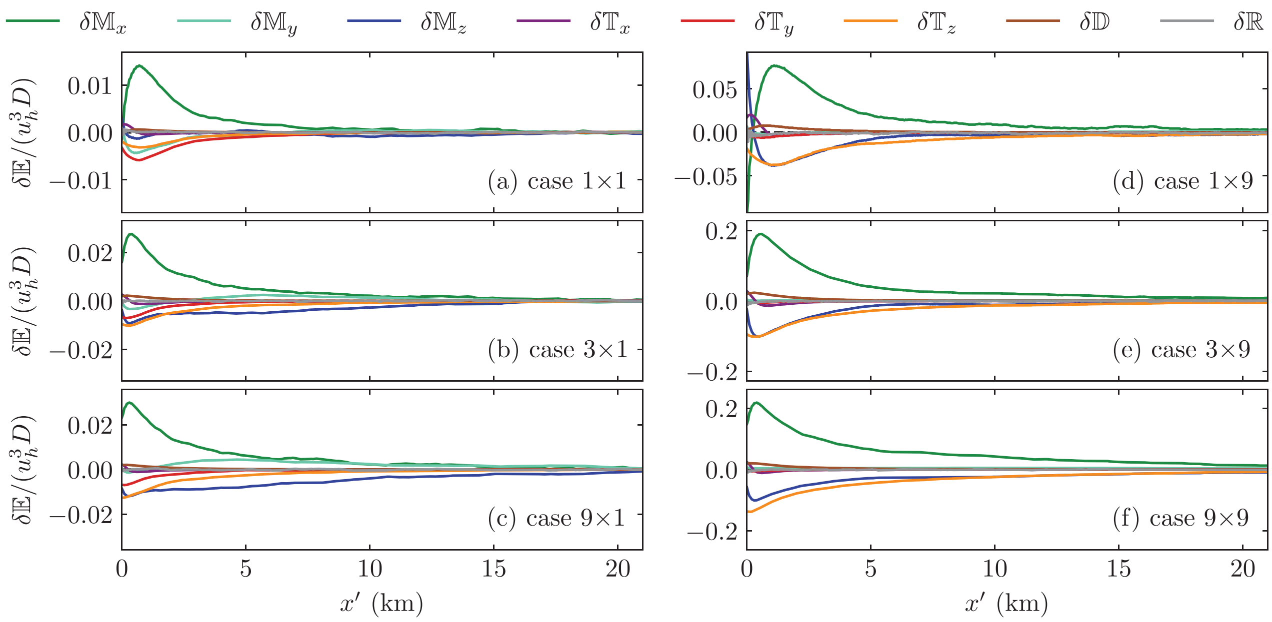

FIG. 9. Overview of the different source and sink terms $\delta\mathbb{E}$, as a function of distance x' behind the wind farm. (a)–(c) Several single-column configurations, and (d)–(f) several 9-column configurations. Positive $\delta\mathbb{M}_x$ indicates wake recovery. The other terms [refer to Eq. (8)] are either sources enabling wake recovery (negative terms), or sinks hindering wake recovery (positive terms).

balance of the precursor flow (a horizontally homogeneous equilibrium between geostrophic forcing, turbulent dissipation, and vertical turbulent transport) is linearly subtracted from the successor domain budget. In this way, we isolate the effects of the wind farm on the flow, simplifying the interpretation of the results. Additionally, we find that the terms $\delta\mathbb{G}$ and $\delta\mathbb{B}$ are negligible in our analysis and, therefore, these are omitted for clarity.

First, we are interested in how the various source and sink terms $\delta\mathbb{E}$ vary with distance downstream. Figure 9 shows $\delta\mathbb{E}$ as a function of x' for several of the simulations. Note that we have expanded the mechanical energy flux and turbulent transport terms in their directional components. The power generation of free-flow turbines is solely driven by the streamwise mechanical energy flux impacting them from the front.[8,41] Hence, a positive $\delta\mathbb{M}_x$ signifies wake recovery; the energy in the flow available for extraction is being replenished. For all simulations $\delta\mathbb{M}_x$ peaks near the exit of the wind farm, before gradually decreasing with distance downstream, indicating a fast near-farm recovery and slower recovery in the far field. The remaining terms from Eq. (8) are either energy sources enabling recovery (the negative terms) or sinks hindering recovery (the positive terms). In most cases, the wake recovery is primarily enabled by energy entering/leaving the wake region due to turbulent mixing with the surrounding flow, either through the sides ($\delta\mathbb{T}_y$) or the top/bottom ($\delta\mathbb{T}_z$). Far downstream, all source and sink terms tend to 0, indicating the equilibrium state of the undisturbed ABL is approached.

Figure 9 reveals that the physics of turbine wake recovery vary significantly from those of wind farm wake recovery. For the larger wind farms, $\delta\mathbb{T}_z$ is the largest energy source replenishing the mechanical energy of the wake, in agreement with findings by Bastankhah *et al.*[48] Moreover, an additional energy source exists in the form of $\delta\mathbb{M}_z$, which represents the mechanical energy entering or leaving the wake region due to the mean vertical flow. In contrast, $\delta\mathbb{T}_y$ is the most dominant energy source in the single-turbine simulation, followed by the spanwise mechanical energy flux $\delta\mathbb{M}_y$. Additionally, $\delta\mathbb{T}_z$ is still significant, though $\delta\mathbb{M}_z$ is small for this case. These findings are consistent with the results of Cortina *et al.*,[7] who found spanwise mass and energy transport to be larger than vertical transport for isolated turbines in neutral ABLs.

To facilitate a more quantitative comparison of the different recovery mechanisms between all simulations, we integrate all source and sink terms $\delta\mathbb{E}$ over the wake length L_{wake},

$$\mathbb{E} = \int_{L_{\text{wake}}} \delta\mathbb{E}(x')\mathrm{d}x, \tag{9}$$

and display the resulting terms $\mathbb{E}$ in a single bar chart in Fig. 10(a). Here, $\mathbb{M}_x$ equals the total power required to raise the streamwise mechanical energy flux back up from its value immediately behind the wind farm to the value at $x' = L_{\text{wake}}$. Logically, $\mathbb{M}_x$ increases significantly with both N_T and M_T, since both increase the size of the wake.

Figure 10(b) displays a normalized version of the graph, which compares the relative importance of the different sources and sinks for all cases. In this graph, we can identify a clear distinction between turbine-scale recovery physics and wind farm-scale recovery physics. For the isolated turbine wake, shaded in red in Fig. 10, $\mathbb{T}_y$ is the dominant energy source, and $\mathbb{M}_z$ is of small magnitude, as we observed in Fig. 9(a). A transition in flow physics is observed for single turbine rows or columns, shaded in blue in Fig. 10. Increasing either the number of rows or the number of columns relative to the 1×1 case, progressively reduces the relative contribution of $\mathbb{T}_y$ and boosts that of $\mathbb{M}_z$. Additionally, there is a significant initial increase in $\mathbb{T}_z$ for both the 3×1 or 1×3 configurations relative to the single-turbine case,

FIG. 10. Comparison of the total source and sink terms $\mathbb{E}$ among all simulated wind farms. (a) Normalized by the hub-height velocity u_h and turbine diameter D, and (b) normalized by the sum of all sink terms $\sum \mathbb{E}_{\rm pos}$. Highlighted are different regimes of wake recovery physics: turbine wake recovery (red), transition (blue), and wind farm wake recovery (white).

though the transition to an even longer row or column has limited impact. Finally, there is a wind farm-scale recovery regime, comprising all the wind farms with at least three rows and three columns of turbines. For these configurations, $\mathbb{T}_z$ is typically the dominant energy source, closely followed by $\mathbb{M}_z$. Spanwise recovery mechanisms have become mostly negligible. The relative importance of the different mechanisms is nearly independent of the exact number of rows and columns in the wind farm in this regime.

So far, we have established that the transition from turbine-scale to farm-scale recovery is primarily characterized by two trends: a reduction in the turbulent entrainment ratio $\mathbb{T}_y/\mathbb{T}_z$, and an increase in the vertical mechanical energy flux $\mathbb{M}_z$. Therefore, we examine these two aspects in more detail below.

A. Turbulent entrainment

The reduction in $\mathbb{T}_y/\mathbb{T}_z$ with increasing N_T and M_T can be understood as the combined effect of two key factors:

(i) for wider wind farms, the aspect ratio of the wake cross section $A_{\rm cs}$ increases, and a larger surface area is available for vertical entrainment relative to spanwise entrainment;
(ii) vertical turbulent entrainment into the wake region remains efficient over larger length scales than spanwise entrainment.

The first factor is easily illustrated. For the single-turbine case, the wake region has a width and height both equal to D; a wake-aspect ratio of 1. Equal surface area exists for spanwise and vertical entrainment. For the 1×3 configuration, the wake width already increases to $L_{\rm WF,y} = (1 + 2s_y)D$; an aspect ratio of 11. Thus, a significantly larger surface area is available for vertical entrainment.

The second factor is revealed as we evaluate the mean flux density $\Delta\mathbb{E}$ (the mean entrainment per unit area) for the different mechanisms at the wake region boundaries; refer to Figs. 8(b) and 8(c). For example, for the vertical turbulent entrainment through the top of the wake region, the flux density $\Delta\mathbb{T}_{z^+}$ is defined as

$$\Delta\mathbb{T}_{z^+} = \frac{1}{L_{\rm wake} L_{\rm WF,y}} \int_{L_{\rm WF,y}} \int_{L_{\rm wake}} A dx dy, \tag{10}$$

where integrand A is given by

$$A = \left(\frac{1}{2} \overline{u_i' u_i' u_3'} + \overline{u}_i \overline{u_i' u_3'} + \overline{u_i \tau_{i3}} + \overline{u_3' p'} \right) \Bigg|_{z = z_h + D/2}. \tag{11}$$

Similarly, the corresponding flux density $\Delta\mathbb{T}_{z^-}$ through the bottom of the wake region requires the integrand to be evaluated instead at $z = z_h - D/2$, and so on. Figure 11 compares $\Delta\mathbb{E}$ for the turbulent recovery mechanisms for all simulations. In the single-turbine case, the total entrainment through the sides $\Delta\mathbb{T}_{y^+} + \Delta\mathbb{T}_{y^-}$ is significantly more efficient than the total vertical entrainment $\Delta\mathbb{T}_{z^+} + \Delta\mathbb{T}_{z^-}$. We

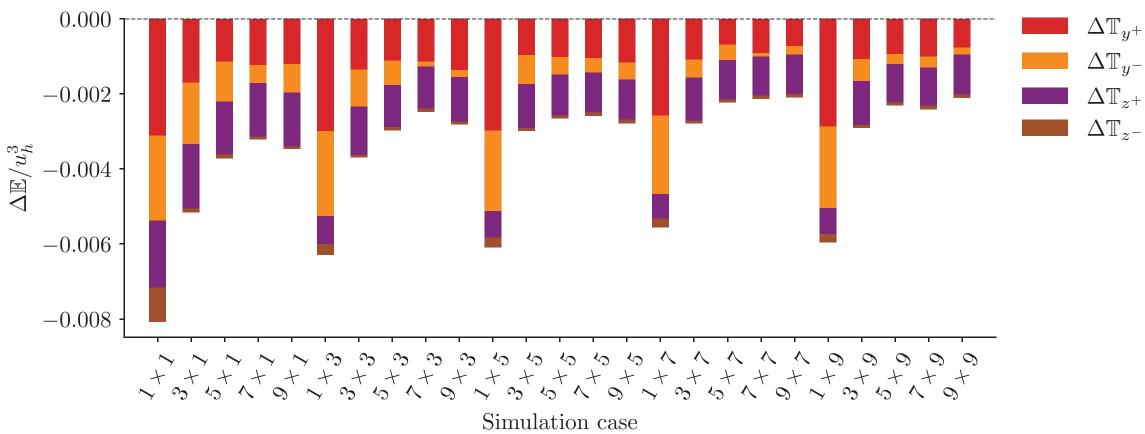

FIG. 11. The mean flux densities $\Delta\mathbb{E}$ at the wake region boundaries for the turbulent recovery mechanisms, shown for all wind farm configurations ($N_T \times M_T$); see also Figs. 8(b) and 8(c). Negative fluxes are directed inward toward the control volume.

re-emphasize here that the energy analysis is performed relative to the undisturbed flow. $\Delta\mathbb{T}_{z^-}$ acts as an energy source not because energy enters the wake region from below, but because the energy leaving the region is reduced relative to undisturbed conditions. As N_T increases, $\Delta\mathbb{T}_{y^-}$ and $\Delta\mathbb{T}_{y^+}$ decrease quickly, while $\Delta\mathbb{T}_{z^+}$ remains fairly constant. A likely cause for this difference is the sustained development of the IBL for the larger wakes. As the wind farm wake expands vertically, it accesses increasingly higher velocity (and therefore higher energy) layers of the atmosphere, which re-energize the flow at lower altitudes. Consequently, vertical entrainment around $z = z_h + D/2$ remains relatively efficient even at large distances downstream. In the spanwise direction, no such development occurs, and the entrainment diminishes more quickly with distance downstream. Finally, we observe that the left and right side entrainment terms, $\Delta\mathbb{T}_{y^+}$ and $\Delta\mathbb{T}_{y^-}$, typically differ in magnitude, in particular for the larger wind farm wakes. This difference results from the asymmetry of the ABL structure, induced by the Coriolis force, which is known to deflect large wind farm wakes.[36]

B. Mechanical energy flux

To understand the increased importance of $\mathbb{M}_z$ in wind farm wake recovery relative to turbine wake recovery, we evaluate the mean vertical flow in the wake region. Figure 12 displays the mean vertical velocity $\langle\overline{w}\rangle$ in the $x'z$-plane for three of the simulations. For the $N_T \geq 3$ cases, turbine wakes accumulate within the wind farm (refer to Fig. 4), and a notable streamwise mass flow deficit develops relative to undisturbed flow conditions. In the wake region behind the wind farm, turbulent entrainment from above accelerates the flow, again increasing the mass flow rate with x', which requires an influx of mass from outside of the wake region to maintain mass continuity. This mass flux is primarily facilitated by the high-energy flow layer aloft in the atmosphere, which has been accelerated by wind farm blockage. Consequently, the induced vertical flow [visualized in Figs. 12(b) and 12(c)] corresponds to a significant mechanical energy flux $\mathbb{M}_z$, which itself further accelerates the flow in the wake region. In extreme cases of long ($N_T \geq 5$) but narrow ($M_T \leq 3$) farms, the mean vertical flow is sufficiently strong that a corresponding outward-directed mass flux

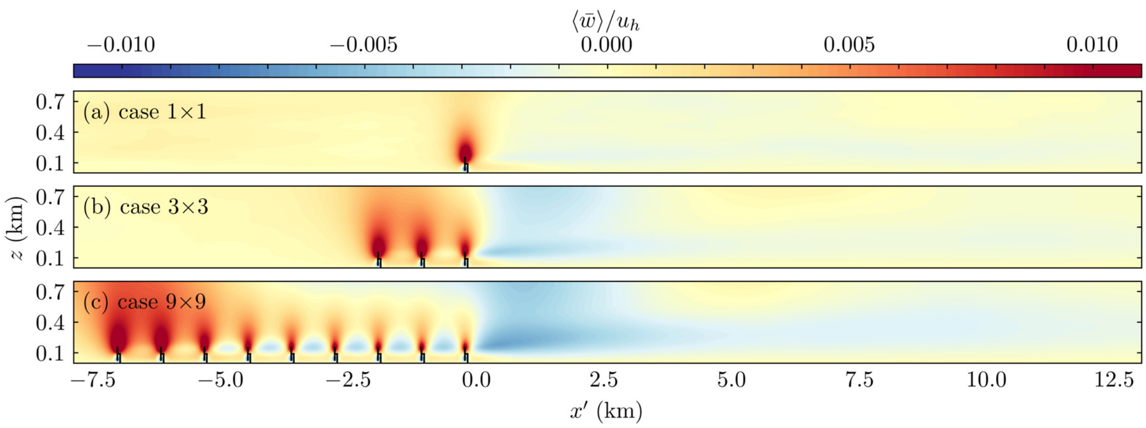

FIG. 12. Time- and spanwise-averaged vertical velocity $\langle\overline{w}\rangle$ in the $x'z$-plane for three of the simulations. Note that only part of the simulation domain is shown, and that the vertical extent is stretched by a factor of two to improve clarity.

exists in spanwise direction. In general, this interaction between $\mathbb{T}_z$ and $\mathbb{M}_z$ helps explain why wind farm wake recovery is much more efficient in deep ABLs compared to shallow ones.[45]

In contrast to the larger wind farm cases, no significant vertical flow develops in the single-turbine simulation, as shown in Fig. 12(a). Instead, the mass flux facilitating wake recovery primarily comes through the sides of the wake region.[7] However, the initial mass flow deficit is relatively small in the first place, and mechanical energy fluxes play a less significant role in the energy transport into and out of the wake, as revealed by Fig. 10.

The importance of $\mathbb{M}_z$ in the wake recovery behind wind farms also highlights a fundamental difference with the wake development within wind farms. Within a large wind farm, past the entrance region, an equilibrium state is often approached, where the energy taken out by the turbines is (almost) entirely compensated via vertical turbulent transport.[4,6] As a consequence, the velocity and mass flow rate remain relatively constant with distance downstream and no global vertical mean flow is induced. In contrast, behind the wind farm turbulent transport accelerates the flow, increasing the mass flow rate and thereby inducing a notable downward mean flow into the wake. Consequently, the energy transport into the wake region is governed by large-scale mechanical energy fluxes in addition to turbulent entrainment fluxes.

In summary, Sec. VI shows a clear difference between turbine-scale and wind farm-scale wake recovery physics. This contrast primarily results from a shift in the relative importance of vertical vs spanwise recovery mechanisms. Our results imply that semi-infinite models may already describe the wake recovery behind wind farms as small as 3×3 turbines, provided they accurately capture the relevant vertical mechanisms. Finally, we note that these conclusions are obtained for a deep CNBL, and other atmospheric conditions can change the wake recovery mechanisms,[45] and potentially the transition between different recovery regimes.

VII. ENGINEERING MODEL EVALUATION

In this section, we compare the mean wake recovery predicted by the engineering models discussed in Sec. II B to our LES. Figure 13

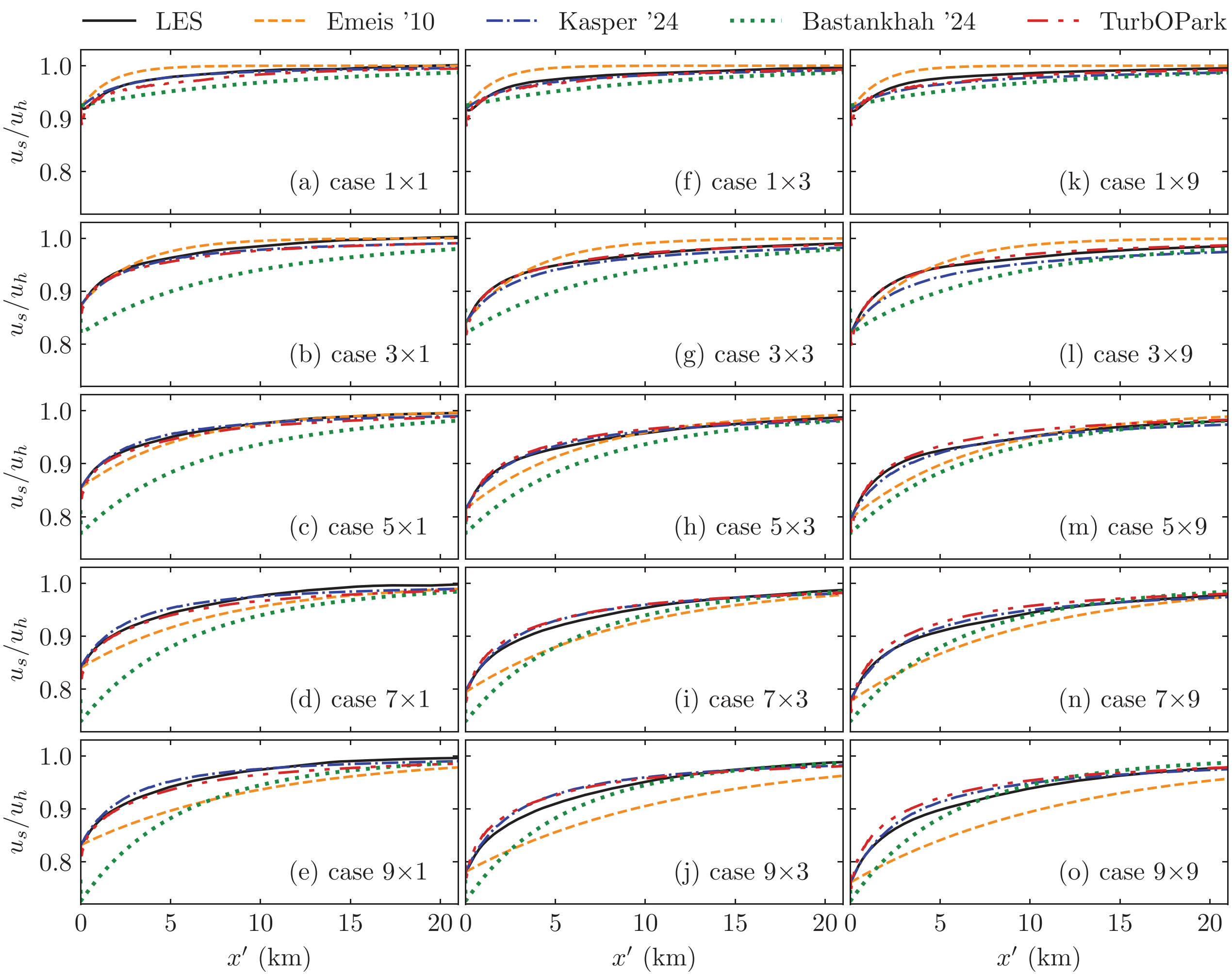

FIG. 13. Comparison between LES and model results of the normalized mean velocity magnitude u_s/u_h at hub height, as a function of streamwise distance x', for different-sized wind farms. (a)–(e) The $M_T = 1$ cases, (f)–(j) the $M_T = 3$ cases, and (k)–(o) the $M_T = 9$ cases.

displays the mean velocity at hub height downstream of the wind farms for 15 of the simulations. Each of the models requires input for the ABL and the wind farm properties, which is provided by the LES data.

The Emeis (2010) model is the simplest of the four models considered, and arguably provides the least consistent results. For the shorter wind farms ($N_T \leq 3$), the recovery rate is overpredicted, while it is underpredicted for the longer wind farms ($N_T \geq 7$), especially in the near-farm wake region. Although the model was designed for spanwise-infinite wind farms, accuracy does not improve with increasing M_T, contrary to expectations. The model's recovery rate parameter α, given by Eq. (B3), is highly sensitive to z_{IBL}, which it receives from the LES data. Our findings demonstrate this equation does not accurately represent the relationship between the IBL development and wake recovery rate found in LES.

In contrast, the Kasper (2024) model yields reliable predictions for the majority of combinations of N_T and M_T considered. The only notable exceptions are the 1×9 and 3×9 simulations, where the recovery is marginally underpredicted in both the near and far wake. Even though the model's intended use is primarily for larger wind farms, it performs well even for the single-column configurations. This likely follows from the fact that the model is based on the same mass-conserving wake expansion mechanism [refer to Eqs. (B4)–(B6)] that numerous turbine wake models use as well, including the TurbOPark model.[54,79,80] This mechanism accounts for both vertical and spanwise turbulent entrainment. It should be noted that the Kasper (2024) model, like the Emeis (2010) model, requires the IBL height and velocity at the farm exit to be known *a priori*, as the flow inside the wind farm is not modeled. Ideally, the model should be extended with estimates for u_{end} and z_{IBL}, to be fully predictive.

The Bastankhah (2024) model appears best suited for both long and wide wind farm configurations. Note that for this model the wind farm exit velocity is modeled, and not required as input, in contrast to the Emeis (2010) and Kasper (2024) models. Given that the Bastankhah (2024) model is spanwise-infinite, the recovery curves depicted in Fig. 13 are independent of M_T. Consequently, we observe that the velocity deficit at $x' = 0$ is considerably overpredicted for the very narrow wind farms, but becomes increasingly more accurate with increasing M_T, as spanwise entrainment is unaccounted for. Furthermore, the model generally underpredicts the near-farm recovery rate, especially for the shorter farms, though yields fairly accurate velocity predictions in the far field (for $M_T \geq 3$). These findings are consistent with the model evaluation of Ref. 48 and indicate that possibly the relative importance of the wind farm and atmospheric contributions could be further optimized.

The TurbOPark model produces similar recovery predictions to the Kasper (2024) model, providing accurate results for a wide range of N_T and M_T. This similarity is not surprising, as both models share a core wake-expansion mechanism. Incorporating both spanwise and vertical entrainment, the TurbOPark model performs well for small to moderate-sized wind farms. The model's capability to accurately predict the farm exit velocity is particularly noteworthy. We find the TurbOPark model starts to overpredict the wake recovery rate with increasing wind farm length and width, which is most evident for the largest (9×9) configuration. This discrepancy likely follows from the model's inability to represent larger-scale interactions with the atmosphere. Furthermore, this model is more costly computationally than the other models presented, as each turbine is parameterized separately.

In summary, no single model performs universally better than the others, but the TurbOPark and Kasper (2024) models provide the most robust predictions within the investigated parameter space. In this context, it is noteworthy that none of the models explicitly account for mean flow energy transport into the wake, which Fig. 10 revealed to be a significant factor in wind farm wake recovery. Moreover, the exact relation between mean flow and turbulent transport—even though the two mechanisms are strongly correlated—can vary with wind farm configuration. How engineering models might benefit from including mean-flow effects remains an open question for future research.

VIII. WIND FARM WAKE STRUCTURE

Until now, we have solely investigated the spatially-averaged properties and recovery mechanisms of the wind farm wakes. Next, we evaluate how the wake structure varies in both the spanwise and vertical directions. Figure 14 depicts the time-averaged velocity in the wake as a function of spanwise distance y at different distances behind the wind farm, for the 3-column and 9-column simulations, respectively. Directly downstream of the wind farm, at $x' = 1$ km, we observe that the characteristics of the individual turbine wakes are still clearly distinguishable within the wind farm wake. For larger N_T, significantly more wake merging has occurred within the wind farm, and consequently, the spanwise variations are reduced. Interestingly, the minimum velocity within the wake does not decrease with increasing N_T. In fact, it is the relatively higher-speed regions within the wake that slow down for larger N_T, therefore resulting in a lower mean wake velocity. We find that the individual wake structures diminishes rapidly with distance downstream for the $N_T \geq 3$ farms. Similar observations have been reported in previous studies.[18,44] Furthermore, we observe that the spanwise wake shape depends strongly on the number of columns in the wind farm. For the configurations with $M_T = 9$, the wake exhibits a top-hat shape, with a plateau of constant velocity in the center. For the narrower $M_T = 3$ wind farms, such a plateau is not observed. That is to say, the wake edges significantly affect the spanwise wake structure in narrow wind farms, but have a lesser impact in wide wind farms.

As revealed already in Fig. 5, wind farms can affect the atmosphere at much higher elevations than hub height, due to the large-scale vertical energy entrainment inside the IBL.[41] Figure 15(a) shows in solid lines the vertical velocity deficit profiles $\delta = u_{\text{ABL}} - u_s$ at different distances x' downstream of the wind farms. The dashed lines illustrate how the height of the velocity deficit maximum $z_{\max}$ develops with x'. For all simulations, $z_{\max}(x' = 0)$ is roughly equal to z_h, as the turbine thrust forces act around this height. Furthermore, we observe that $z_{\max}$ shifts upwards with increasing distance downstream. This effect is expected and enabled by the prevailing wind shear in the ABL.[41,49] For the isolated turbine, the wake recovers relatively fast and the upward shift is small; $z_{\max}$ does not exceed the physical height of the turbine. For the larger wind farms, the increased velocity deficit and turbulence intensity inside the wind farm (shown in Fig. 4) intensify the development of the IBL. Given the large depth of the ABL, the IBL has space to develop unhindered. As a result, we find that the upward shift of $z_{\max}$ increases with wind farm size, persisting further downstream.

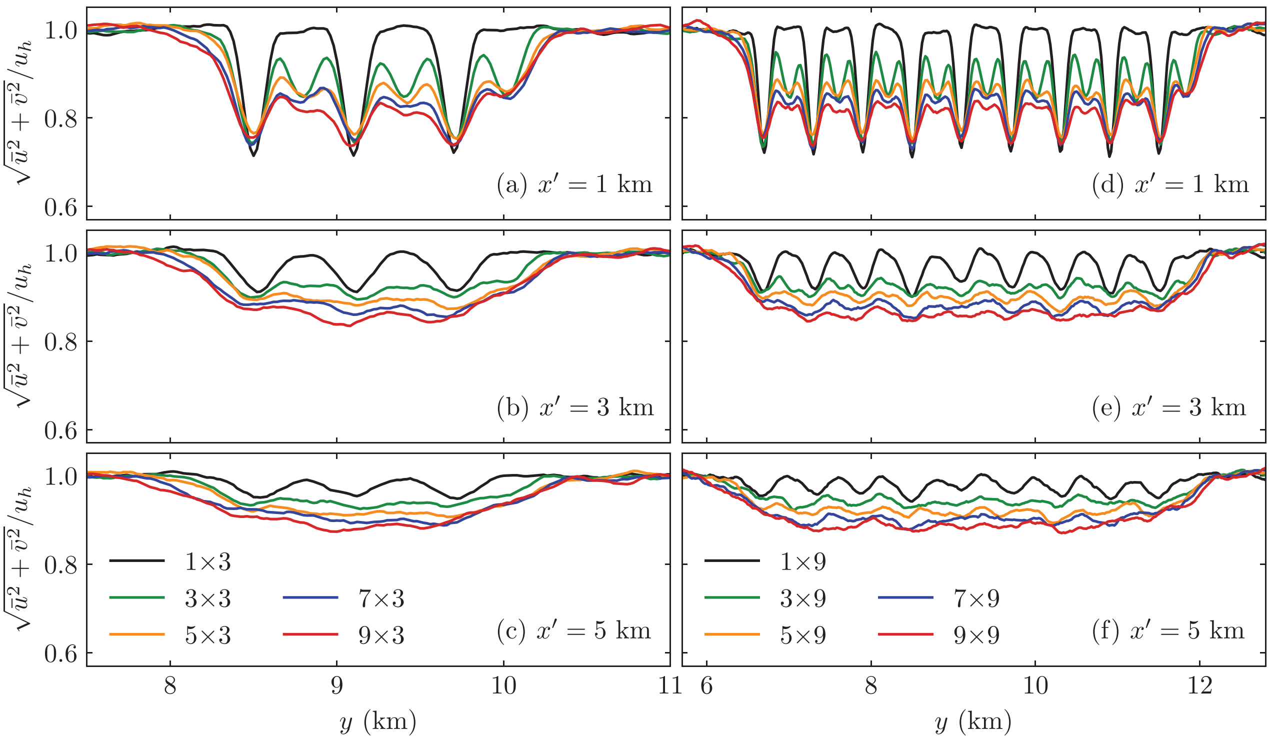

FIG. 14. Time-averaged spanwise velocity distributions at hub height, at different distances x' downstream of the wind farm. (a)–(c) The $M_T = 3$ cases and (d)–(f) the $M_T = 9$ cases.

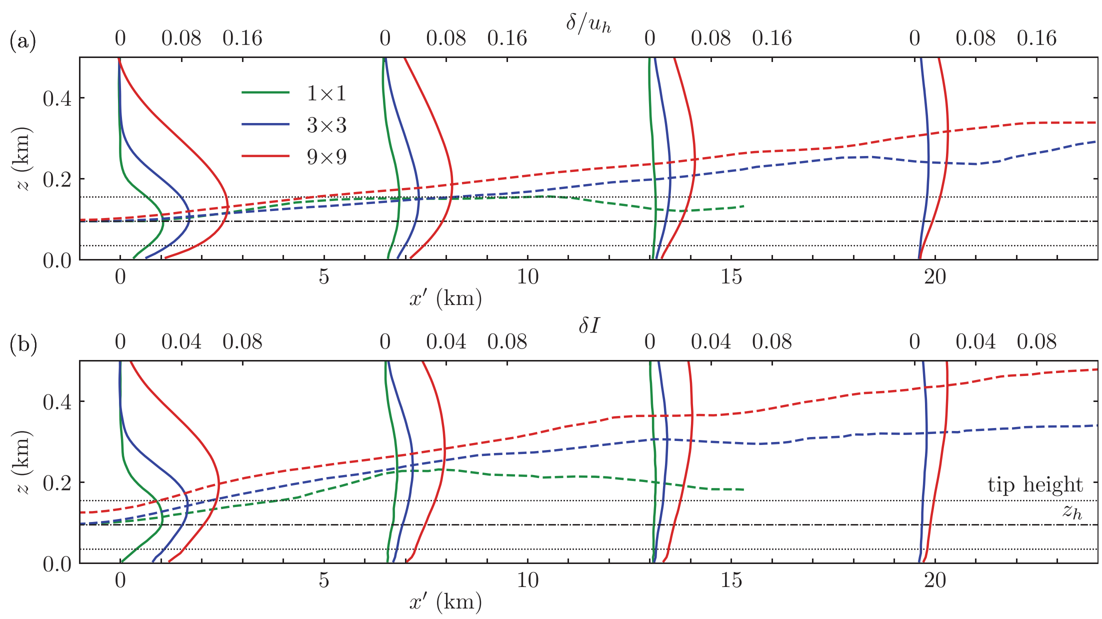

FIG. 15. Development of wake properties with height z and distance x' downstream of the wind farm, for three of the simulations. (a) The normalized velocity deficit δ/u_h and (b) the turbulence intensity surplus δI. The solid lines depict vertical profiles at different distances downstream. The dashed lines depict the evolution of the height z_{max} where the deficit/surplus is largest. The horizontal black dash-dotted and dotted lines denote hub height z_h and the turbine rotor height, respectively. The 1×1 line is cut off where the wake has fully recovered (and z_{max} becomes ill-defined).

The trends observed for the velocity deficit δ are also reflected by the surplus in turbulence intensity $\delta I = \langle I \rangle - I_{\mathrm{ABL}}$. Figure 15(b) shows the development of δI downstream of the 1×1, 3×3, and 9×9 wind farms. We observe that at $x' = 0$, the additional turbulence intensity is highest around hub height, but $z_{\max}$ increases with x', especially for the larger wind farms. Interestingly, the upward shift is more pronounced for δI than for δ / u_h, and is even observed within the wind farm already for the 9×9 case. In summary, the results of Fig. 15 demonstrate that larger wind farms exhibit an increased capability to alter the atmosphere at higher elevations far downstream of the farm, despite the fact that at hub height the wake is already (mostly) recovered.

IX. CONCLUSION

We performed large eddy simulations (LES) to study the wake recovery downstream of 25 wind farm configurations, ranging from 1 to 81 turbines, in a deep neutral boundary layer. We evaluated the mean flow properties, recovery mechanisms and structure of the wind farm wakes, while incrementally exploring the effects of additional rows and columns, and found significant dependencies on both the length and width of the wind farm. Moreover, our analysis reveals several key insights that improve our physical understanding of wake recovery, in particular on the differences in recovery mechanisms between turbine-scale and wind farm-scale wake recovery. These insights can be leveraged to improve the tools used to site wind farm within clusters.

The velocity development of a wind farm wake is characterized by a fast near-farm recovery, and a slower far-field recovery. Furthermore, if the wind farm is large enough to form a well-mixed aggregated wake, the far-field recovery becomes independent of the configuration of the wind farm. The length of the wake increases with both the number of rows N_T and columns M_T in the wind farm; however, the rate of increase diminishes as more turbines are added. Specifically, the wake length plateaus around $M_T \geq 7$ in our simulations. Based on these findings, we determined that the wind farm wake length follows an asymptotic exponential decay function, though *a priori* prediction of its coefficients proves non-trivial. Nonetheless, this functional relationship may be useful in practical applications, as it enables simple extrapolation of wake properties to wind farms of different sizes.

Furthermore, we evaluated several engineering models designed to predict large-scale wake recovery. Although each of these models is limited in their representation of wake-atmosphere interaction, several of them yielded fairly accurate predictions of the mean-velocity recovery at hub height. In particular, the Kasper (2024) model[41] and the TurbOPark model[54] provided good agreement with the LES for a wide range of wind farm sizes. Both these models accurately represent the near-farm and far-field recovery and capture both spanwise and vertical entrainment processes.

By performing an analysis of the wake recovery mechanisms, we demonstrate that the physics of turbine-scale and wind farm-scale wake recovery differ significantly. Behind an isolated turbine, wake recovery is driven mainly by spanwise turbulent entrainment, with smaller contributions from spanwise mechanical energy fluxes and vertical turbulent transport. In contrast, for wind farms comprising at least three rows and three columns of turbines, vertical turbulent transport and vertical mechanical energy fluxes are the dominant terms, while spanwise recovery mechanisms are negligible. Smaller wind farms fall into a transition regime and exhibit a combination of both the turbine and wind farm wake recovery physics. We identified two key factors that govern the shift from spanwise to vertical recovery physics. First, the wake cross section aspect ratio increases with wind farm width, thereby increasing the surface area available for vertical (relative to spanwise) energy fluxes. Second, vertical turbulent entrainment remains efficient over larger length scales than spanwise entrainment. The role of the vertical mechanical energy flux in farm-scale recovery—created by the downward flow downstream of the wind farm—is noteworthy, since energy transport by the mean vertical flow is often considered unimportant within large wind farms.

Multiple topics for future research emerge from the findings presented in this study. Foremost, these findings were obtained for wind farms in a deep conventionally neutral boundary layer. Further investigations are required to study how conditions such as thermal stability, boundary layer height, mesoscale phenomena and complex terrain impact wind farm wake recovery physics. Furthermore, future research should expand to even larger (and in particular, longer) wind farms, to evaluate whether an infinite-wind farm regime is attained,[6,13] and how this relates to the development of the internal boundary layer, which can be limited by the capping inversion height.[43] Additionally, gravity waves are known to significantly affect wind farm wake development in shallow ABLs, introducing strong oscillations in multiple key recovery terms.[45] To what extent and how this affects the transition between turbine-scale and farm-scale recovery still remains unknown. Moreover, the research presented here could be extended to wakes downstream of clusters of wind farms, to uncover if these exhibit even larger-scale recovery physics. Finally, future work should examine how the presented findings on wake recovery, including mean flow effects, can guide improvements to engineering wake recovery models.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

J. H. Kasper: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal); Writing – review & editing (equal). **R. J. A. M. Stevens:** Conceptualization (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Resources (equal); Software (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX A: VALIDATION OF THE GRID RESOLUTION

To validate that the employed grid resolution in our work of $\Delta_x = 50$ m, $\Delta_y = 20$ m, and $\Delta_z = 10$ m is sufficiently fine to capture the wakes and recovery mechanisms, we compare the single-turbine case to a higher resolution simulation. This higher resolution LES is performed on a smaller domain of $L_x = 13$ km by $L_y = 4.8$ km, with a doubled horizontal grid resolution of $\Delta_x = 25$ m and $\Delta_y = 10$ m.

Figure 16 displays the time-averaged wake centerline velocity u_w as a function of distance x' behind the turbine for both simulations. We find there is good agreement between both curves, apart for the near-wake region immediately downstream of the turbine, where the coarse-grid simulation slightly underpredicts the wake deficit. However, the coarse-grid curve collapses onto the fine-grid curve within a short distance downstream. At the virtual location of the subsequent turbine row (at $s_x = 7D$), the velocity difference is below 2%.

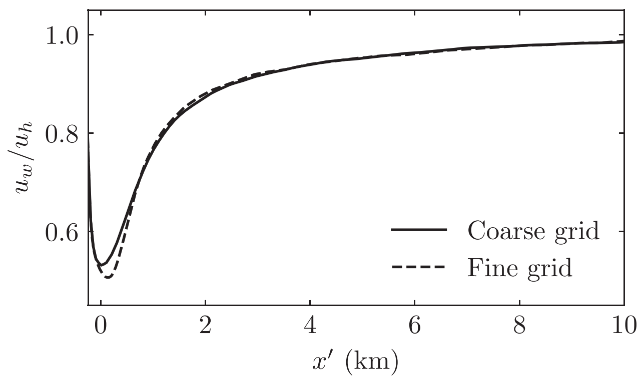

FIG. 16. Time-averaged wake centerline velocity u_w/u_h as a function of distance x', compared between the coarse- and fine-grid LES.

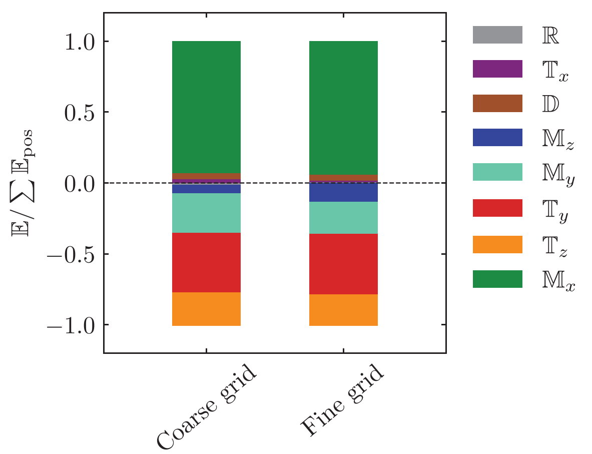

FIG. 17. Comparison of the normalized energy source and sink terms $\mathbb{E}$ in the wake region. Refer to Fig. 10 and Eq. (8) for variable definitions.

Figure 17 compares between both simulations the energy sources and sinks in the turbine wake region, as defined in Sec. VI. We find that the grid resolution only has a minor influence on the relative importance of the different budget terms, and the major recovery mechanisms are captured relatively well even by the coarse grid. Therefore, the employed grid resolution is sufficient to study the turbine-scale to wind farm-scale wake recovery transition. It should be noted, however, that in studies where quantitative assessment of the near-wake region is of greater interest, a finer grid should ideally be used to ensure that the local wind speed deficit and velocity gradients are captured with sufficient accuracy.

APPENDIX B: MATHEMATICAL DESCRIPTION OF THE ENGINEERING MODELS

1. Emeis (2010) model

Following Ref. 53, let $\delta = u_h - u$ be the wind velocity deficit behind the wind farm, with undisturbed hub-height velocity u_h and local velocity u. For an infinitely wide wind farm, it is assumed δ is with time entirely compensated by the vertical momentum flux τ_{ver}, i.e.,

$$\frac{\partial u}{\partial t} = -\frac{\partial \tau_{\text{ver}}}{\partial z}. \tag{B1}$$

The momentum flux is considered to be driven by the vertical wind shear as $\tau_{\text{ver}} = K_m \partial u/\partial z$, where the momentum flux coefficient is estimated as $K_m = \kappa u_* z_{\text{IBL}}$, with internal boundary layer height z_{IBL}. Then, approximating several derivatives using differences, and employing the approximation $t = x'/u_h$ to convert temporal units into spatial ones, one finds a solution to the above differential equation of the form

$$\delta(x') = \delta_0 \exp(-\alpha x'). \tag{B2}$$

Here, δ_0 is the initial velocity deficit immediately behind the wind farm, and x' denotes the distance behind the wind farm. Furthermore, the wake expansion is governed by the parameter

$$\alpha = \frac{K_m}{u_h (z_{\text{IBL}} - z_h)^2}, \tag{B3}$$

where z_h denotes hub height.

In summary, Eqs. (B2) and (B3) make up the full model, where z_{IBL} and δ_0 must be known or estimated for the considered wind farm. Furthermore, the model uses the friction velocity u_* of the undisturbed ABL.

2. Kasper (2024) model

Fundamentally, the Kasper (2024) model[41] follows a similar mass conservation argument as first introduced in the models of Refs. 79 and 80. The velocity deficit $\delta(z, x')$ at height z and distance x' behind a wind farm is given by

$$\delta(x',z) = g(x')\delta_0 \frac{Z_{w0}Y_{w0}}{Z_w(x')Y_w(x')} \exp\left\{\frac{-[z-z_c(x')]^2}{2\sigma^2(x')}\right\}. \tag{B4}$$

Here, $Z_{w0} = 2D$ and $Y_{w0} = y_{\mathrm{WF}}$ are the estimated initial wake height and width, respectively, where y_{WF} is the width of the wind farm itself. The local wake height Z_w and wake width Y_w at some distance x' behind the wind farm are estimated using

$$Z_w(x') = Z_{w0} + 2kx', \tag{B5}$$

$$Y_w(x') = Y_{w0} + 2kx', \tag{B6}$$

where the wake expansion parameter $k = 0.45 I_{\mathrm{ABL}}(z_c)$ depends on the atmospheric turbulence intensity I_{ABL} at the local wake center height z_c. The wake has a Gaussian shape in the vertical, with $\sigma(x') = Z_w(x')/5.16$, such that 99% of the velocity deficit lies within the wake height. Furthermore, the model considers an upward shift of the wake center height based on the vertical atmospheric wind shear, empirically described by

$$z_c(x') = z_h + 0.08 \frac{u_{\mathrm{ABL}}(z_{\mathrm{IBL}}) - u_h}{u_h} x'. \tag{B7}$$

Here, $u_{\mathrm{ABL}}(z_{\mathrm{IBL}})$ denotes the undisturbed wind velocity at the IBL height [note that $u_h = u_{\mathrm{ABL}}(z_h)$]. The ground effect is accounted for by correction factor $g(x')$, which preserves the total mass flow deficit for $z \geq 0$.

Similarly to the Emeis (2010) model, z_{IBL} and δ_0 must be known or estimated for the considered wind farm. Furthermore, the model uses the velocity and turbulence intensity profiles $u_{\mathrm{ABL}}(z)$ and $I_{\mathrm{ABL}}(z)$ as input conditions.

3. Bastankhah (2024) model

As detailed in Ref. 48, for a turbine row n, the streamwise and spanwise velocity deficits $\delta_{U,n}$ and $\delta_{V,n}$ as a function of distance x'_n behind the turbine row are given by

$$\delta_{U,n}(x'_n) = A\cos(-f_c x'_n/u_h)u_h, \tag{B8}$$

$$\delta_{V,n}(x'_n) = A\sin(-f_c x'_n/u_h)u_h. \tag{B9}$$

Here, A is defined as

$$A = \frac{C_T}{2s_y}\left(1 - \eta_n \frac{\delta_{\mathrm{in},n}}{u_h}\right)^2 \exp\left(-\frac{c_1}{u_h D^2}\int_0^{x'_n} \nu_t \mathrm{d}x'_n\right), \tag{B10}$$

where $c_1 = 0.85$ is an empirical coefficient and $\delta_{\mathrm{in},n}$ represents the local inflow to row n. Furthermore, η_n is the local deficit coefficient, which accounts for the turbine layout (e.g., staggered or aligned configuration). $\nu_t = \nu_{t,f} + \nu_{t,0}$ is the total turbulent viscosity, which governs how quickly the wake recovers. It comprises an ambient component $\nu_{t,0} = \kappa u_* z_h$, and a wind farm component $\nu_{t,f}$, which depends on the roughness length z_0, the ABL height z_{ABL}, various turbine/farm properties and an empirical coefficient $c_2 = 0.05$; refer to Ref. 48.

The total deficits $\delta_{U,n}$ and $\delta_{V,n}$ are then found by summing the contributions of the individual rows and adding the atmospheric corrections

$$\delta_U(x) = \frac{C_x D^2}{c_1 \nu_t}\left[1 - \exp\left(-\frac{c_1}{u_h D^2}\int_0^x \nu_t \mathrm{d}x\right)\right] + \sum_n \delta_{U,n}(x), \tag{B11}$$

$$\delta_V(x) = \frac{C_y D^2}{c_1 \nu_t}\left[1 - \exp\left(-\frac{c_1}{u_h D^2}\int_0^x \nu_t \mathrm{d}x\right)\right] + \sum_n \delta_{V,n}(x). \tag{B12}$$

Here, the groups of terms containing C_x and C_y represent the wind shear and veer, respectively. C_x and C_y are determined using the geostrophic drag law, the turbulent viscosity, and an empirical constant $c_3 = 0.06$, further detailed in Ref. 48.

In summary, the model provides physics-based wind farm flow predictions on a row-by-row basis, using the atmospheric properties u_*, z_0, z_{ABL}, and f_c.

4. TurbOPark model

In the TurbOPark model,[54] the velocity deficit δ_n of an individual turbine n is given as a function of distance x'_n behind the turbine by

$$\delta_n(x'_n) = \delta_{0,n}\left[\frac{D}{D_w(x'_n)}\right]^2. \tag{B13}$$

To estimate the wake diameter $D_w(x'_n)$, first the turbulence intensity at x'_n is estimated as $I_{\mathrm{loc}}(x'_n) = \sqrt{I_w^2(x'_n) + I_{\mathrm{ABL}}^2}$, where the wake-generated turbulence is described as

$$I_w(x'_n) = \frac{1}{c_1 + c_2 x'_n/(D\sqrt{C_T})}. \tag{B14}$$

Here, $c_1 = 1.5$ and $c_2 = 0.8$ are empirical constants. Assuming the wake expansion rate is locally proportional to $I_{\mathrm{loc}}(x'_n)$ yields after integration the following expression for the local wake diameter:

$$D_w = D + \frac{A I_{\mathrm{ABL}} D}{\beta}\left(\sqrt{(\alpha + \beta x'_n/D)^2 + 1} - \sqrt{1+\alpha^2} - \ln\left[\frac{\alpha\left(\sqrt{(\alpha+\beta x'_n/D)^2 + 1} + 1\right)}{(\alpha + \beta x'_n/D)\left(\sqrt{1+\alpha^2}+1\right)}\right]\right), \tag{B15}$$

with $\alpha = c_1 I_{\mathrm{ABL}}$ and $\beta = c_2 I_{\mathrm{ABL}}/\sqrt{C_T}$. Nygaard *et al.*[54] recommend for the wake expansion parameter a value of $A = 0.6$, which we have adopted here as well. The initial deficit $\delta_{0,n}$ in Eq. (B13) is estimated as

$$\delta_{0,n} = u_h - u_{0,n}\sqrt{1 - C_T}, \tag{B16}$$

where $u_{0,n}$ is the disk-averaged inflow velocity for turbine n. In Ref. 54, Nygaard *et al.* couple their wake model to a blockage model, such that both upstream wake effects and downstream blockage effects are considered in the estimation of $u_{0,n}$. In this work, we do not consider such a coupling and only model the wakes.

To model a full wind farm, velocity deficits of individual turbines are added in quadrature,[80] as

$$\delta(x,y,z) = \left[\sum_n \delta_n^2(x,y,z)\right]^{0.5}. \quad \text{(B17)}$$

To account for the ground effect, the wind farm is mirrored below the bottom wall.

In summary, Eqs. (B13) and (B15)–(B17) make up the model, which leverages insight into the atmospheric turbulence intensity I_{ABL} at hub height.

APPENDIX C: NUANCES IN SPANWISE AVERAGING OF WIND FARM FLOW PROPERTIES

The exact definition of what is the "spanwise extent" of the wind farm over which averages are taken, can influence the perceived effect of the number of turbine columns M_T on the flow properties. Figure 18(a) displays the development of the mean flow u_s/u_h, where the average is taken over the *physical width* $L_{\mathrm{WF},y}$ of the farm: from the leftmost rotor tip to the rightmost. A consequence of this convention is that the turbine density is inconsistent between simulations of varying M_T, particularly for narrower farm configurations. The observed difference in farm exit velocity $u_{\mathrm{end}} = u_s(x' = 0)$ between cases 1×3 and 1×9 is a direct result of this convention. For the longer wind farms, spanwise velocity variations at the farm exit are much smaller, and M_T is of little influence on u_{end}. Furthermore, Fig. 18(a) reveals the interesting observation that the flow blockage directly in front of the farm is affected considerably by M_T: the wider wind farms exhibit a larger flow deceleration in the induction region.

Meanwhile, Fig. 18(c) considers a *fixed width per turbine*, equal to the spanwise spacing $s_y D$, such that the averaging region extends beyond the physical width of the farm on both sides. Using this convention, M_T is perceived to not affect the near-farm wake region in the single-row configurations; the turbine wakes do not merge, so the flow behind each turbine is nearly identical. For larger N_T, wakes start to merge with wakes of neighboring turbines. Consequently, the flow behind the outer turbines behaves differently than the internal ones, as they have neighboring turbines on one side but not the other. These differences are captured using the convention of Fig. 18(c), and the wider wind farms exhibit larger mean velocity deficits as a result. The differences between Figs. 18(a) and 18(c) demonstrate that averaging conventions must be chosen and communicated with care when comparing flows through wind farms of different sizes. The same argumentation holds for the turbulence intensity $\langle I \rangle$, displayed in Figs. 18(b) and 18(d).

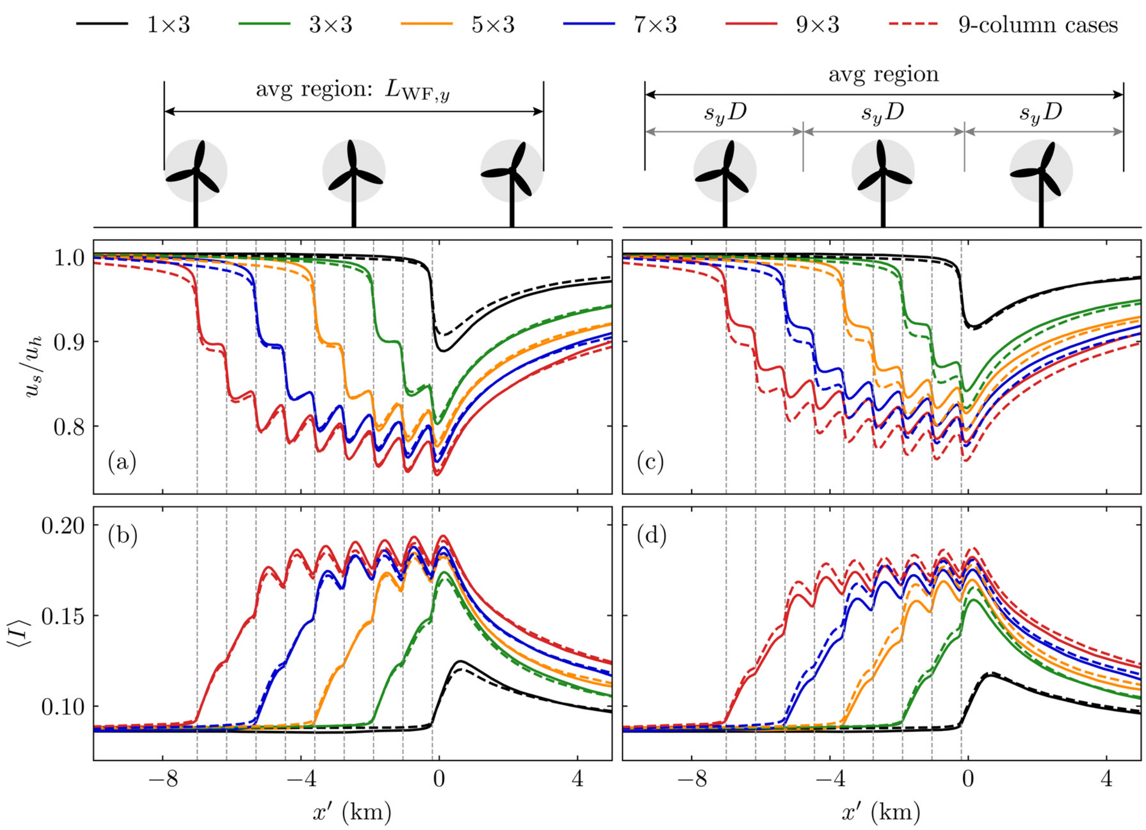

FIG. 18. Temporally- and spanwise-averaged properties inside the wind farm as a function of distance x', at hub height, for the 3- and 9-column cases. (a) and (b) the velocity magnitude u_s and the turbulence intensity $\langle I \rangle$, averaged over $L_{\mathrm{WF},y}$. (c) and (d) u_s and $\langle I \rangle$, averaged over a fixed width $s_y D$ per turbine. The gray dashed lines denote the turbine positions.

TABLE III. Fitting constants for the curves displayed in Fig. 7, constructed using Eq. (7). a denotes the asymptote in km, and b describes how quickly that asymptote is approached.

Curve	a (km)	b
$N_T \times 1$	7.1	0.23
$N_T \times 3$	12.2	0.20
$N_T \times 5$	15.2	0.17
$N_T \times 7$	15.8	0.17
$N_T \times 9$	15.8	0.17
$1 \times M_T$	1.7	2.1
$3 \times M_T$	6.2	0.68
$5 \times M_T$	9.4	0.73
$7 \times M_T$	11.1	0.71
$9 \times M_T$	12.1	0.63

APPENDIX D: FITTING CONSTANTS FOR THE WAKE LENGTH CURVES

Table III documents the fitting constants corresponding to the curves displayed in Fig. 7, which are constructed using Eq. (7). Here, a is the wake length asymptote in km, and b describes how quickly that state is approached as a function of N_T or M_T. We find that for the atmospheric conditions considered in this study, extrapolation of the LES data yields a maximum wake length of 15.8 km. Note that these atmospheric conditions are quite beneficial for wake recovery; the ABL is deep ($z_{\mathrm{ABL}} = 1$ km) and neutrally stratified. For shallow and/or stably stratified ABLs, the maximum wake length is expected to be significantly larger.

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